



# **Systems Engineering - SySe**

## **Chapter 5 – Modeling and Simulation**

Prof. Dr. Silke Lechner-Greite

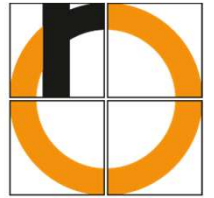
# Learning objectives

- Understand the terms modeling and simulation
- Modelling and Simulation with Product Life Cycle
- Experimental Model
- Simulation Data Management



# Modeling

# What is Modeling?



**Model = Abstraction of a system** (or other topics of interest, i.e., processes, phenomena in general)

**A representation of some real physical item**

The model represents a **set of aspects** of the system, i.e., functionality, performance, geometrical shape, etc.

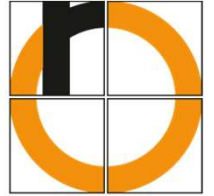
A model represents the system **within its operational environment**

Models are represented by **different model types**, according to the system needs

Models as an **analyzation, specification, design and verification system**

Models as **communication tool** between involved parties of the system development life cycle, with the stakeholders/customers

# What is Simulation?



**Simulation = virtual experiment on a model**, typically computer-aided

Simulation is an **enabler to use the defined models over time**

Simulation is particularly good for **analyzing complex dynamic** behavior of a system.

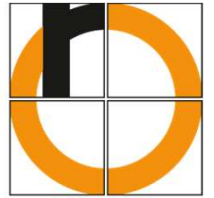
Simulations for **systems, software, hardware, people, physical phenomena** (i.e. flow, thermal properties, etc.)

Simulation represents an analytical model and **considers inputs into the model to predict a certain outcome**.

Simulations include **computing infrastructure** (computational engine, input/output devices)

The simulation needs to provide efficient solving methods → **performance**

# Modeling and simulation ...

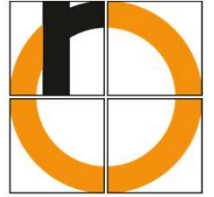


Confirmation - to  
confirm the need  
for the system

Prediction -  
anticipate the  
system behavior

Clarity - clear  
representation  
and design of the  
system

- ... detect system limitations early
  - Safe cost, stay in schedule
  - Models and simulations are performed before actually committing resources in terms of constructing a prototype
- ... important with the size / complexity of the system
- ... supports decision making
  - deliver affordable and timely data during i.e. analyzation
- ... clear representation that helps the communication with customers/stakeholders
  - Good understanding of status that support to re-think alternatives early in the design phase
- ... helps to decide on personnel, methods, tools, and infrastructure for the system



## Why models?



To prevent cost overruns.

überblick der abdeckung von den requirements durch die komponente > dadurch müssen diese nicht später zeit- und kostspielig eingebaut werden/ iterativ überarbeitet werden



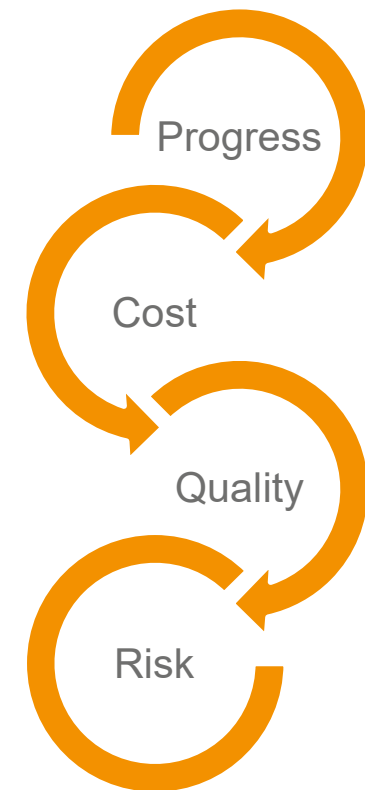
To prevent schedule overruns.



To identify problems early on.

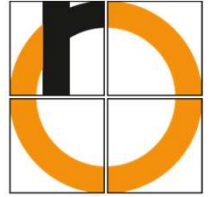
# A model must have a clear purpose!

- **Models for characterizing existing systems:** models help to reproduce a system architecture and design
- **Models support the system concept formulation & evaluation:** evaluate alternatives early in the development state
- **Models support the system architecture design** and formation by considering system requirements
- **Models support system integration and verification** – i.e., when software and hardware is combined into a whole system
- **Models support the training of people** who interact with the system
- **Models as a method to capture knowledge** and system evolution



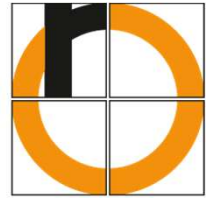


# Model characteristics



- **Model scope**<sup>umfang</sup>: The model used must be tailored to its intended use
- **Model breadth**: degree to which model addresses functional, interface, performance, physical and non-functional system requirements  
wie viele perspektiven werden abgedeckt? isoliert oder zusammenspiel mehrerer komponente?
- **Model depth**: decomposition from overall system context down to individual system elements  
wie tief wird das modell aufgeschachelt?
- **Model fidelity**: level of detail, refers to precision of a model, too.  
detailgrad: wie genau sollen die ergebnisse sein? muss nicht quantitativ/simulativ sein, sondern kann ähnlich zu prototyping zunächst experimentell etwas prüfen
- **Model viewpoints**: function, performance, reliability, accuracy, specification reachability, survivability, operational availability, cost.

# Fitness for Purpose example

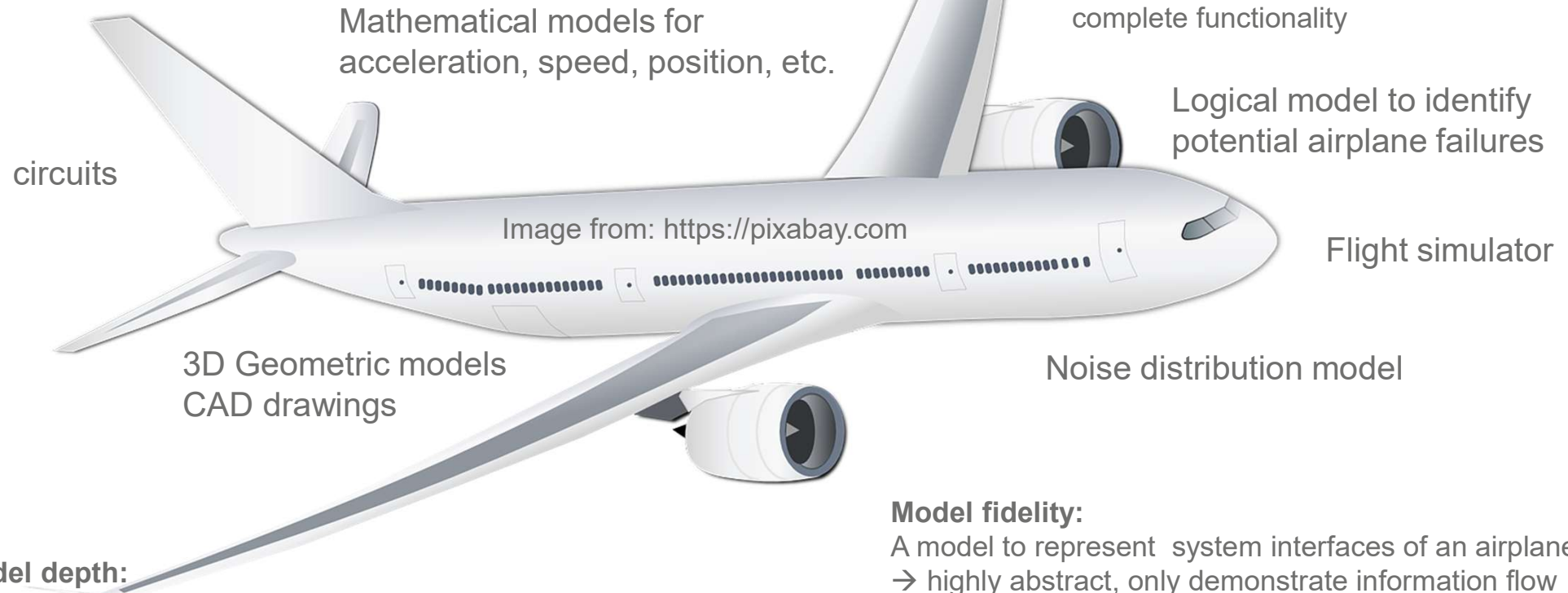


## Model scope:

- System architecture model: interconnection between airplane parts
- Trajectory analysis model: analyze a certain parameter of the airplane over time
- Fault tree analysis: analyze potential reasons for airplane failure

## Model breadth:

Aircraft functional model: if, for example, the model is to be used to model start-up, take-off, flight, landing, shutdown and maintenance, the model can be limited to parts of the functionality, or it can map the complete functionality



## Model depth:

The aircraft models require a system context, i.e., airplane lands at airport controlled via the tower in combination with the airplane navigation tools that can contain finer units such as measurement units.

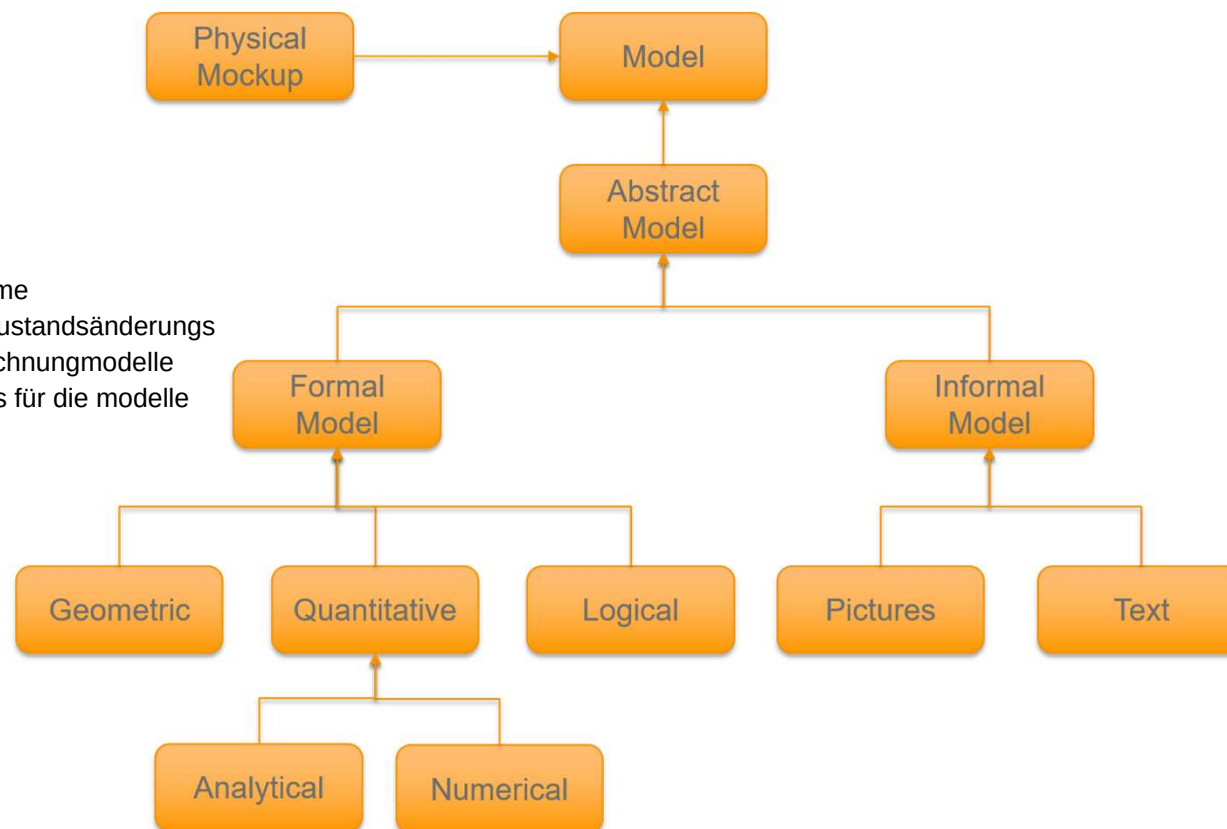
## Model fidelity:

A model to represent system interfaces of an airplane  
→ highly abstract, only demonstrate information flow on aircraft status data

A model to represent how the data is exchanged using a dedicated protocol and encryption contains high detail.

# Model types (cont.)

- Key goal: “Fitness for Purpose”
  - Find the most appropriate model and simulation that represents your system of interest best
  - Select an appropriate level of detail; focus on a subset of characteristics
  - Define the needed accuracy of the model and simulation

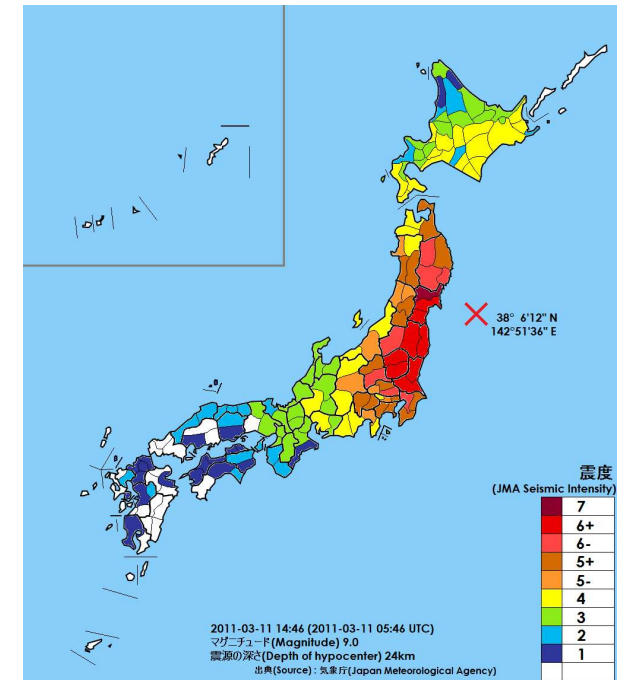


CAD modelle  
theoretische bäume  
mathematische zustandsänderungs  
numerische berechnungsmodelle  
> zahlen als basis für die modelle

A system is modelled with various models within different domains – models for different system components (thermal behavior, vibration behavior), technology domains (HW, SW), characteristics (physical concepts, performance)

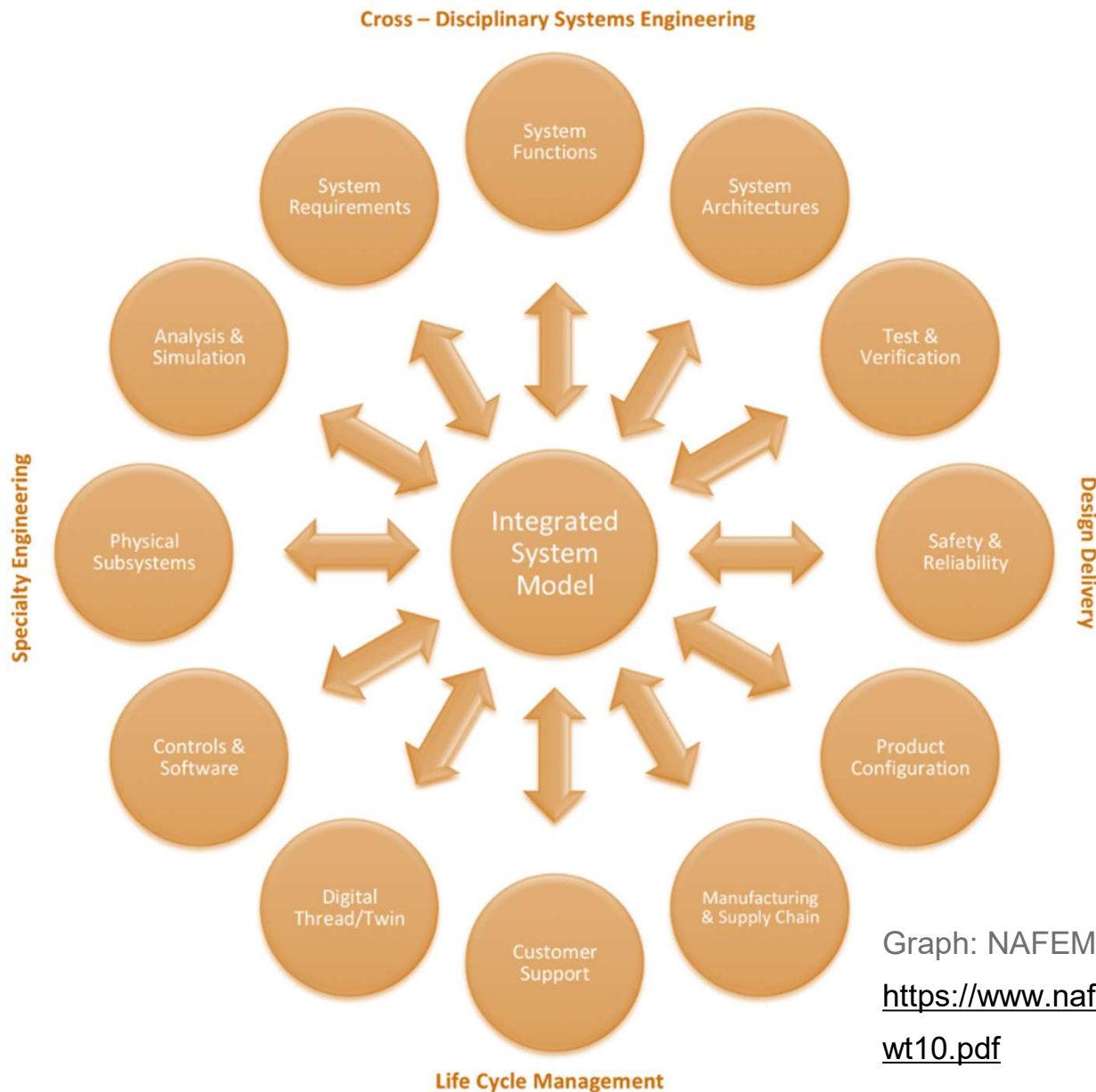
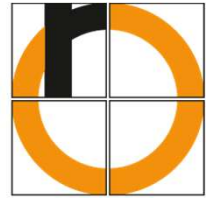
# How to find the “right” model?

- Ask the following questions:
  - **What exactly should be modelled?**
    - The weather forecast of Germany or the weather in Rosenheim?
    - How it is to drive a particular car versus how the car performs ergonomically (wind tunnel)
  - **Which variables play a role (qualitatively) and how great is their influence (quantitatively)?**
    - search for real, measurable parameters and substantiate them scientifically.
    - Deduce hypothesis and assumptions that are vague and speculative which might have a wrong influence on (simulation) result
  - **In what relation are the important variables?**  
and how do they influence one another
  - **What tools can be used to describe phenomena, variables, and their relationships best?**
    - Algebraic equations, inequality, differential equations, partial differential equations
    - Automats, state transition diagrams
    - Graphs – workflows, sequences, states, etc.
    - Neuronal networks



[https://de.wikipedia.org/wiki/Tōhoku-Erdbeben\\_2011#/](https://de.wikipedia.org/wiki/Tōhoku-Erdbeben_2011#/)

# Modeling for complex technical systems

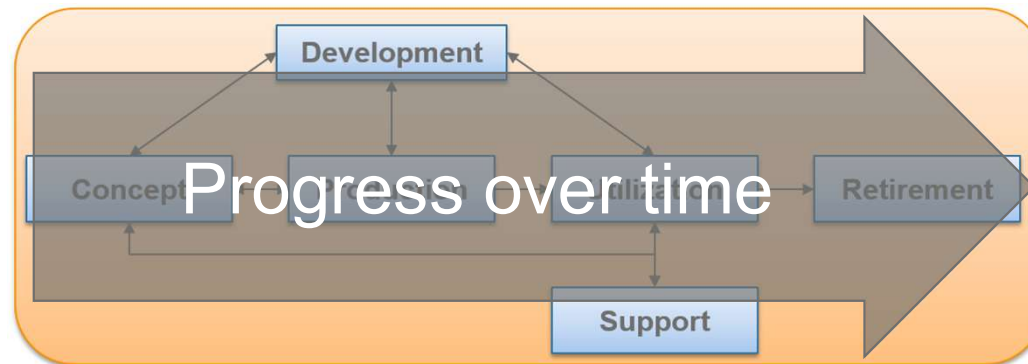
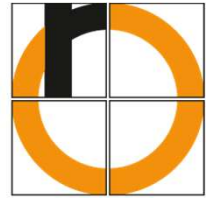


**Keep in mind:** there is rarely the one model, it is rather a model based on the evaluation between effort and accuracy and the perspective one want to represent.

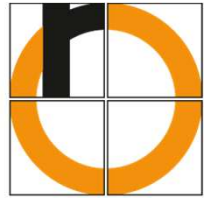
**Like the real system:** it's model representation also touches interdisciplinary fields that are integrated into a model.

Graph: NAFEMS: “What is system modeling and simulation”:  
[https://www.nafems.org/downloads/dropbox/resource\\_center/](https://www.nafems.org/downloads/dropbox/resource_center/wt10.pdf)  
[wt10.pdf](https://www.nafems.org/downloads/dropbox/resource_center/wt10.pdf)

# Verification, Validation & Accreditation



- Modelling and simulations are closely aligned with system life cycle activities
- Planning of models and simulations recommended to achieve low risks.
- Verification and validation of models in parallel to their development is critical as the models will be involved in decisions along the path of the life cycle --- credibility == fit for purpose
- Process to use: Verification, Validation & Accreditation Process
- Key questions:
  - **Verification** – *Did I build the thing right?*
  - **Validation** – *Did I build the right thing?*
  - **Accreditation** – *Is it believable enough to be used?*



## Where to use models?

During requirements analysis phase



During designing the system architecture



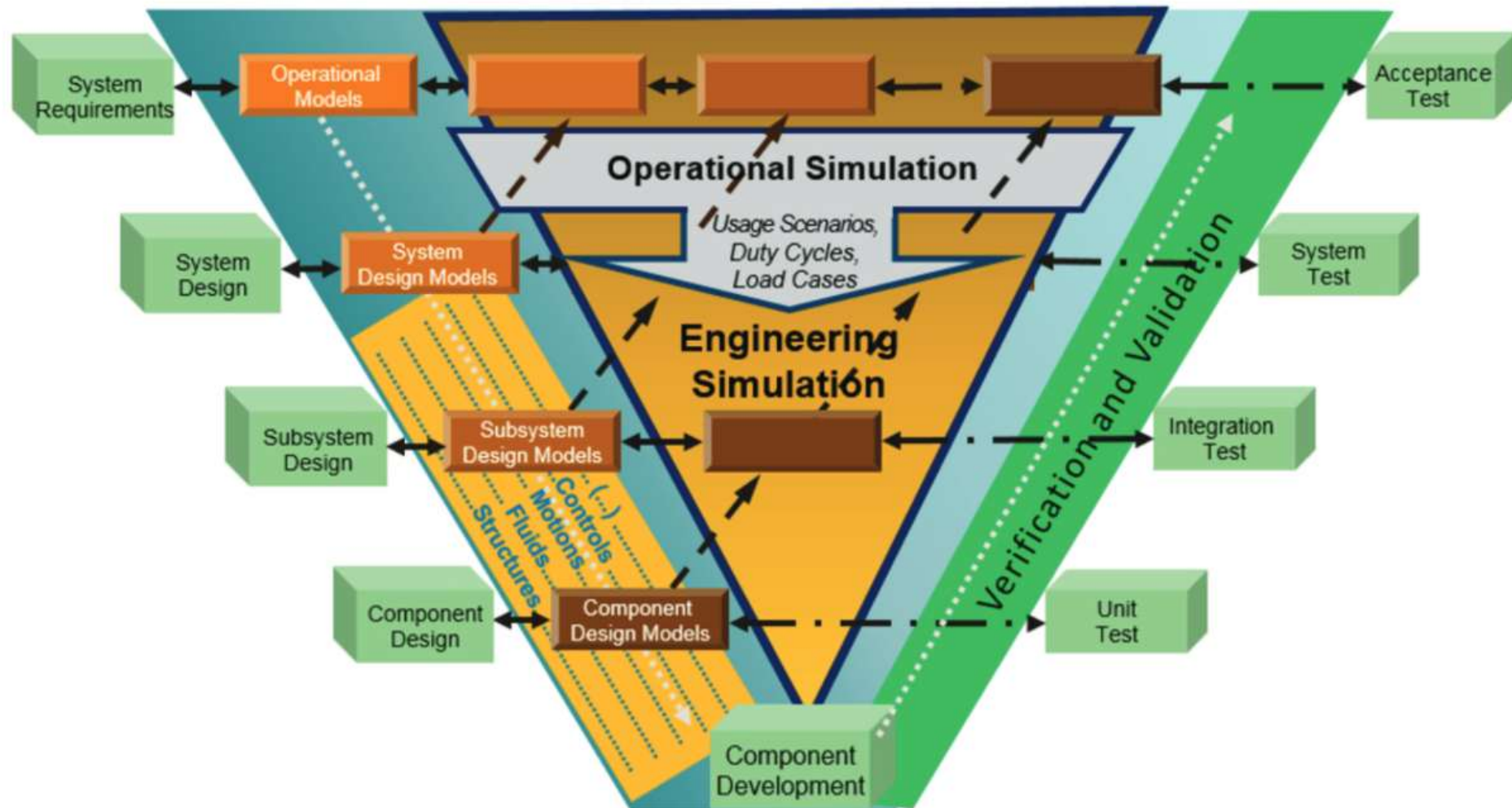
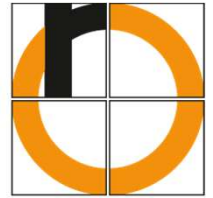
During the development phase



During the operation and maintenance phase.



# Product development supported by simulation



Graph: NAFEMS: "What is system modeling and simulation":

[https://www.nafems.org/downloads/dropbox/resource\\_center/wt10.pdf](https://www.nafems.org/downloads/dropbox/resource_center/wt10.pdf)



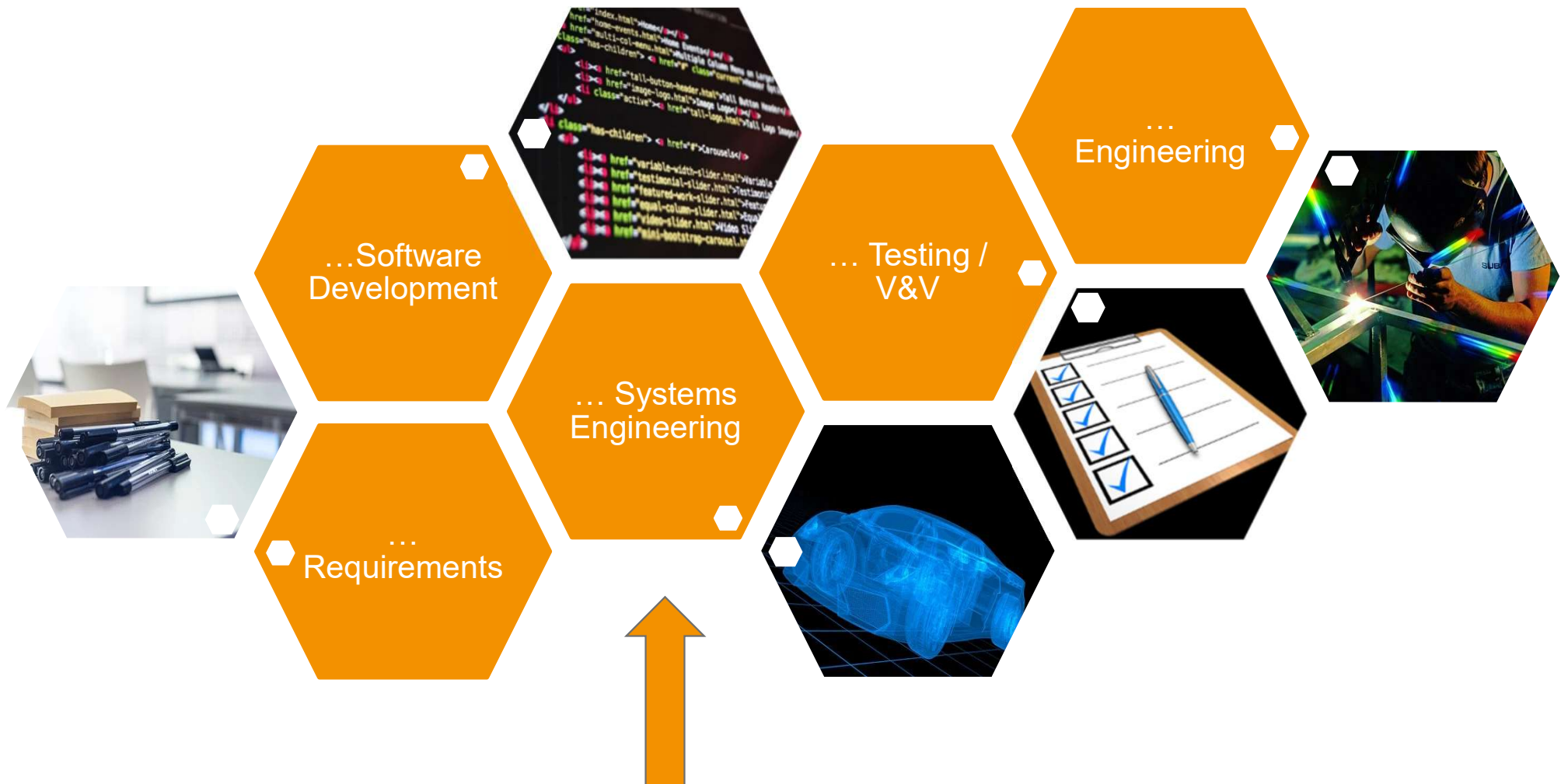
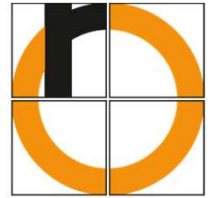


**MBSE**



- Complexity is growing in all dimensions.
- Classic models reach their limits:
  - Products are becoming more and more complex (cyber-physical systems, software integration, KI).
  - Communication between disciplines is becoming more difficult (Mechanics, Electronics, Software, System Architecture).
  - Document-based SE approaches aren't sufficient anymore.
  - Let's look at an approach that transfers the benefits of model-based thinking from modeling and thinking to the entire system development process.

# Model Based ...



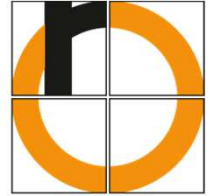
# What is MBSE?

- “ ... *Model-Based System Engineering is the formalized application of modeling to support system requirements, design, analysis, verification and validation activities beginning in the conceptual design phase and continuing throughout development and later life cycle phases. ...*”  
[SE Vision 2020; **INCISE**. 23. Feb. 2009]
- “ ... Model-Based Engineering (MBE): An approach to engineering that uses models as an integrated part of the technical baseline that includes the requirements, analysis, design, implementation, and verification of a capability, system, and/or product throughout the acquisition life cycle. ...”  
[Final Report, Model-Based Engineering Subcommittee, NDIA, Feb. 2011]

Formalized application of modeling

+ models as an integrated part

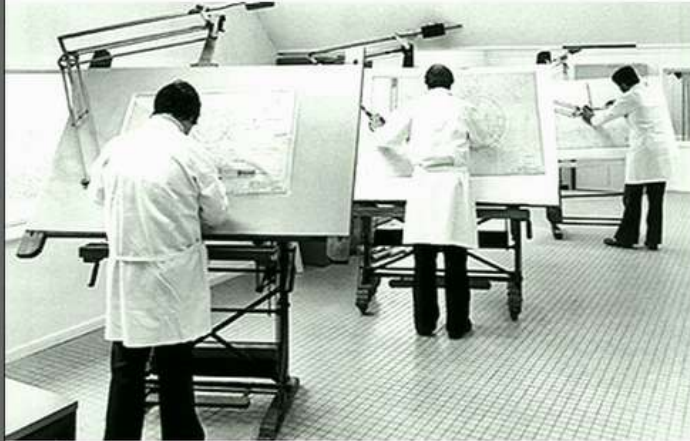
# Idea of MBSE?



 **What's the Big Idea of MBSE?**  
Model-Based Systems Engineering

**Add rigor and precision (enhance communication)**

**Help manage complexity**

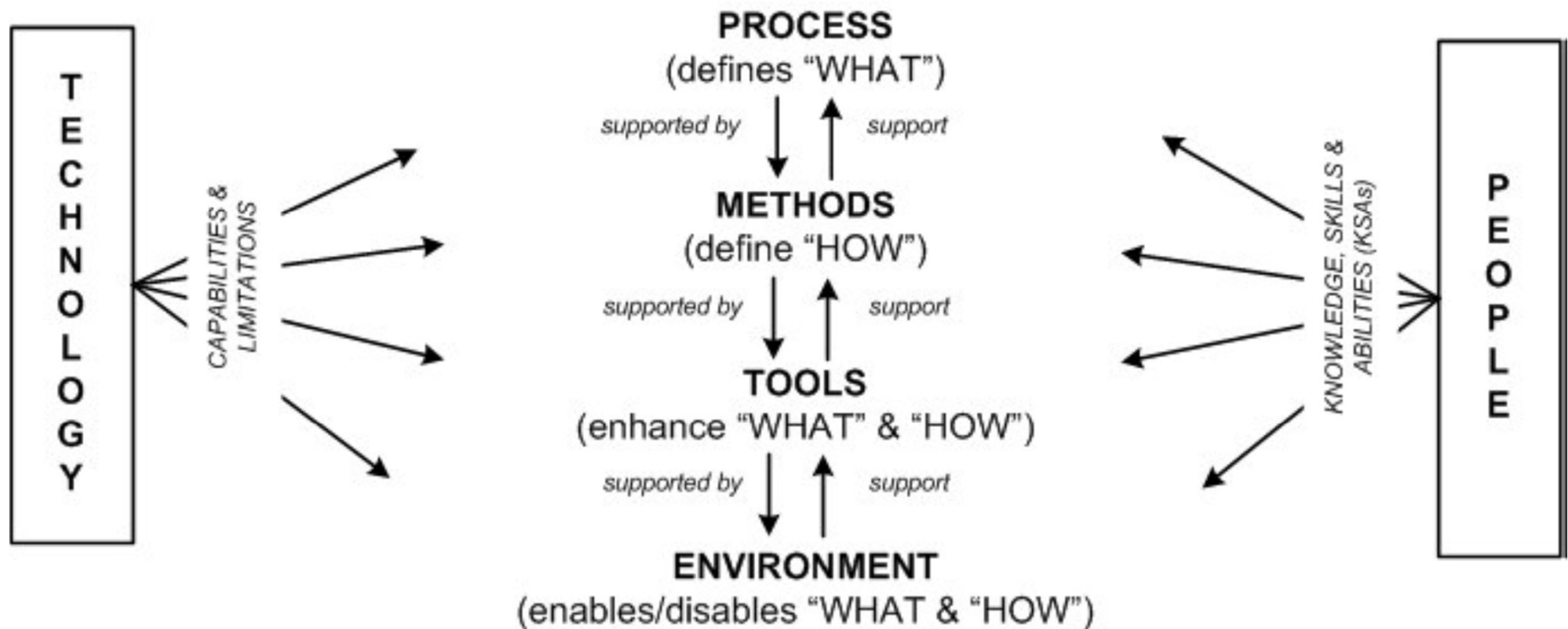
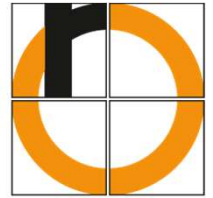
  

Electronic and mechanical CAD went through similar transformations over the last 30 years

Systems Engineering with System Models: An Introduction to MBSE

Screenshot taken  
from the online  
presentation:  
[https://mediaex-  
server.larc.nasa.gov  
/Academy/Play/804f  
524f16c24fd2927a4  
eb9f01bdf831d](https://mediaexchange.larc.nasa.gov/Academy/Play/804f524f16c24fd2927a4eb9f01bdf831d)  
Accessed on  
05.11.2024

# Key Idea of an MBSE Environment

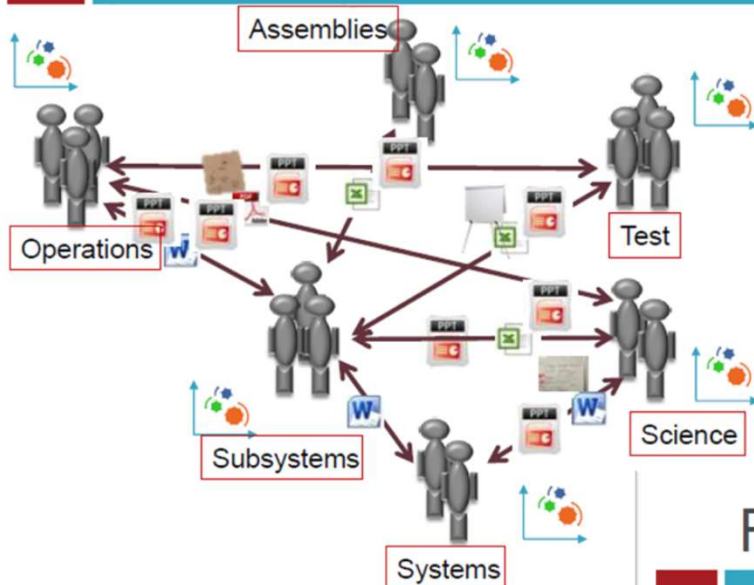


(1) [https://www.youtube.com/watch?time\\_continue=73&v=RaOejcczPas&feature=emb\\_title](https://www.youtube.com/watch?time_continue=73&v=RaOejcczPas&feature=emb_title) (accessed on 26.6.2020) --- unfortunately not available anymore ... (5.11.2024 --- hopefully can recreate the link again)



# Traditional SE vs. MBSE

## Traditional SE Practice



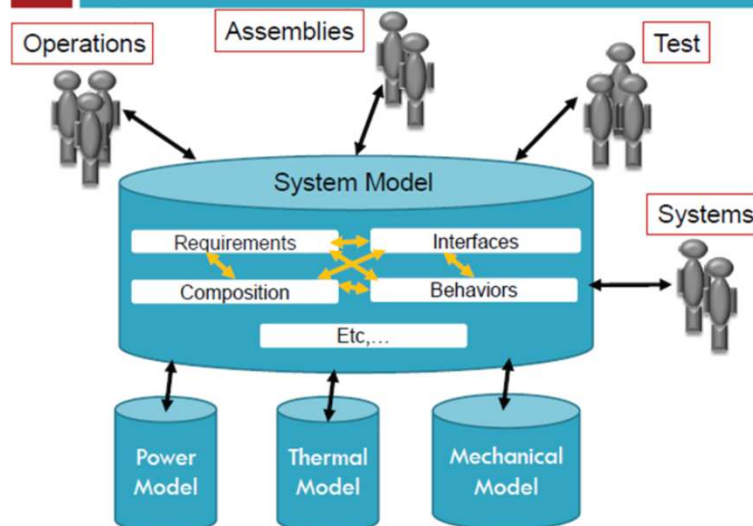
Stand alone domain models/designs related via:

- Institutional life cycle **documents**
  - Operations Concepts
  - Requirements Documents
  - Interface Documents
  - Deployment Plans
- Formal review **presentations**
- **Informal communications**
  - White boards
  - Design Team Meeting Presentations
  - Email
  - Chat
  - Napkin

viele Teams, die eigene Artefakte erstellen  
Austausch findet direkt zwischen den einzelnen Teams statt  
Konstante Updates nötig

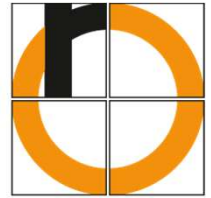
aws, escher

## Future MBSE Practice

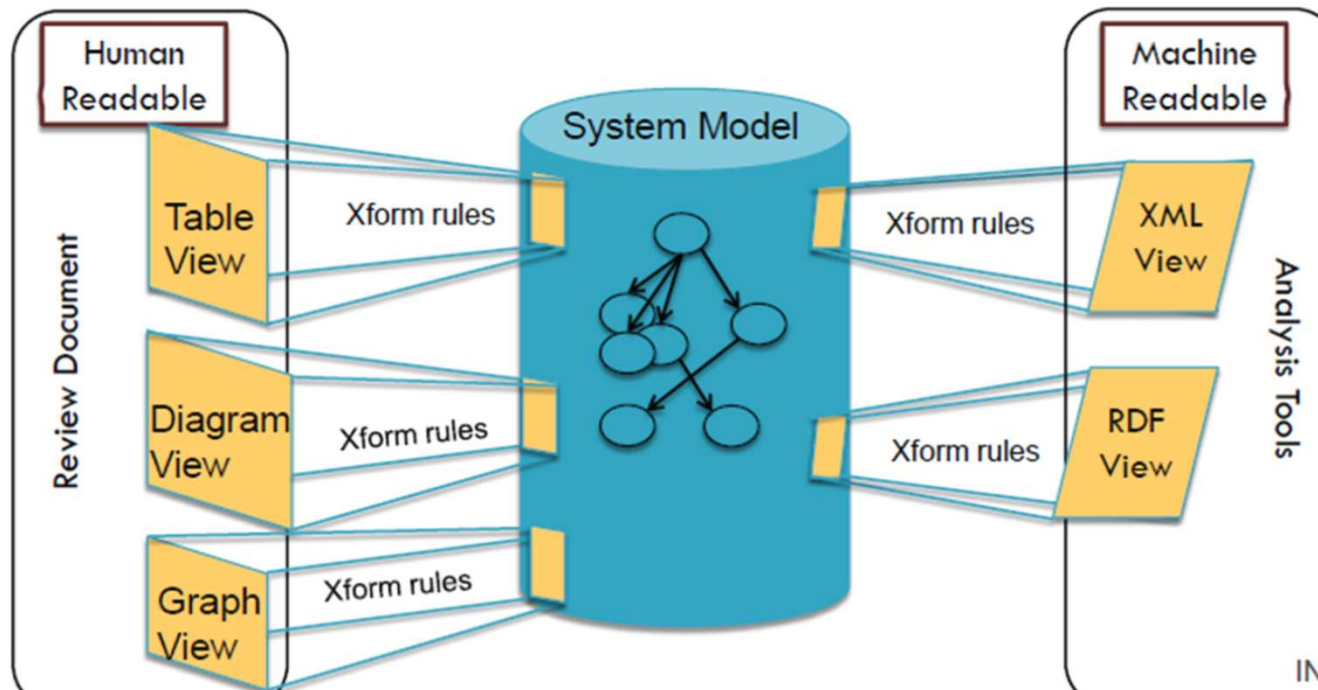


- Integrated system model** with multiple views, connected to discipline models
- **Authoritative source** of information
  - Exchanges information to/from analysis and stakeholders via **projections of model information**
  - Information accessible to all members of the project

Source: Survey of Model-Based Systems Engineering (MBSE) Methodologies by Jeff Estefan, JPL (2008)  
([http://www.omgwiki.org/MBSE/lib/exe/fetch.php?media=mbse:2016\\_iw-mbse\\_101.pdf](http://www.omgwiki.org/MBSE/lib/exe/fetch.php?media=mbse:2016_iw-mbse_101.pdf))  
(downloaded 26.6.2020)



## System Model Projections



- Ability to **interrogate the design information** and extract data into the format necessary for the given task
- Leverages **formalism**
- Transformation rules are **reusable**
- Provides machine and human readable formats

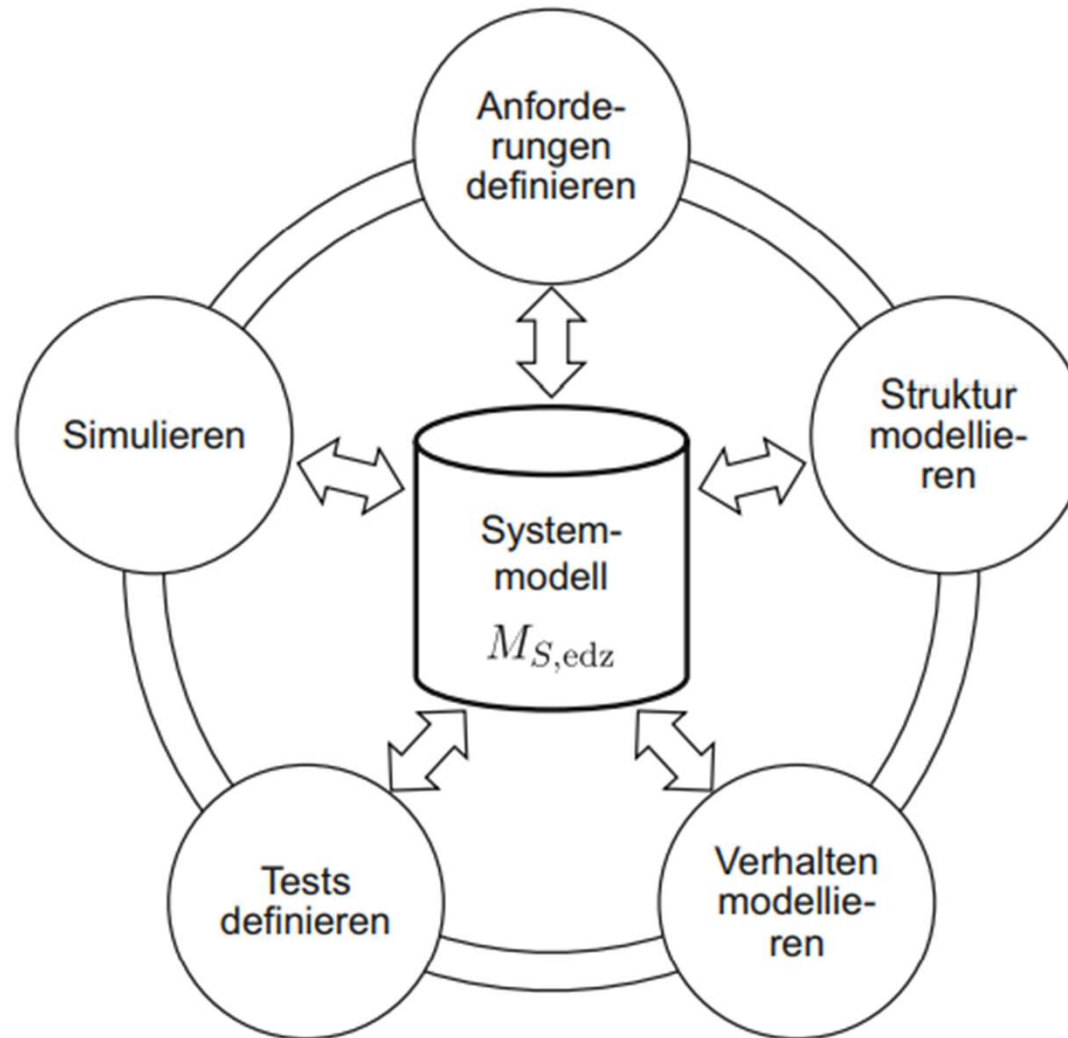
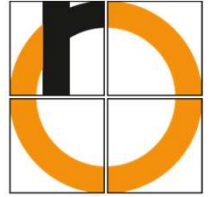
INCOSE IW January 30th, 2016 \* Fosse

Source: Survey of Model-Based Systems Engineering (MBSE) Methodologies by Jeff Estefan, JPL (2008)  
([http://www.omgwiki.org/MBSE/lib/exe/fetch.php?media=mbse:2016\\_iw-mbse\\_101.pdf](http://www.omgwiki.org/MBSE/lib/exe/fetch.php?media=mbse:2016_iw-mbse_101.pdf))  
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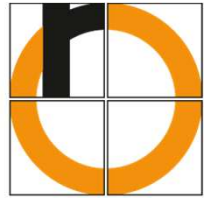
• **System Model either represented in a central model or a federation of models**  
**= “single source of truth”**



# The system model at the heart of MBSE's processes



Source: Schluse et al., Experimentierbare Digitale Zwillinge, Springer, 2024



# MBSE Benefits

**Enhances  
Communication**

**Increases  
Productivity**

**Improves  
Quality**

**Supports  
Integration**

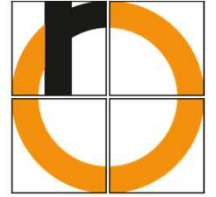
**Manages  
Complexity**

**Facilitates  
Reusability**

Based on : Dave Nichols and Chi Lin. Integrated Model-Centric Engineering: The Application of MBSE at JPL Through the Lifecycle. INCOSE International Workshop, January 26 2014.

disadv: close to single point of failure  
if modell is unavailable or destroyed through wrong input,  
then the whole system comes down

# MBSE still is SE



Dangers

- Assumption that all models are good.
- Models for everything.
- Documenting models only.

models for everything:  
> models with the sole purpose to have a model (informationsschrott) not because it is needed

Philosophy

- Documentation,
- Representation,
- Communication,
- Calculation,
- Building

- Derivation,
- Association,
- Decomposition,
- Dependency

Don't forget:  
MBSE is SE, SE  
is still needed.

Relationships

# Examples of MBSE Methodologies

- Focus on embedded and software-intensive systems: Embedded Computer System Analysis and Modeling (**ECSAM**) methodology:

Jonah Z. Lavi and Joseph Kudish, Systems Modeling and Requirements Specification Using ECSAM: An Analysis Method for Embedded and Computer-Based Systems, Dorset House Publishing Company, Inc.:New York, NY, 2004.

<https://ieeexplore.ieee.org/document/1316676>

- Focus on Software Engineering: Model-Based [System] Architecture and Software Engineering (**MBASE**);

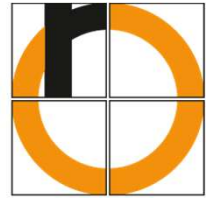
Boehm, Barry and Dan Port, “When Models Collide: Lessons from Software Systems Analysis,” IT Professional, IEEE Computer Society, Institute of Electrical and Electronics Engineers, Vol. 1, Issue 1, Jan/Feb 1999.

<https://ieeexplore.ieee.org/document/774794>

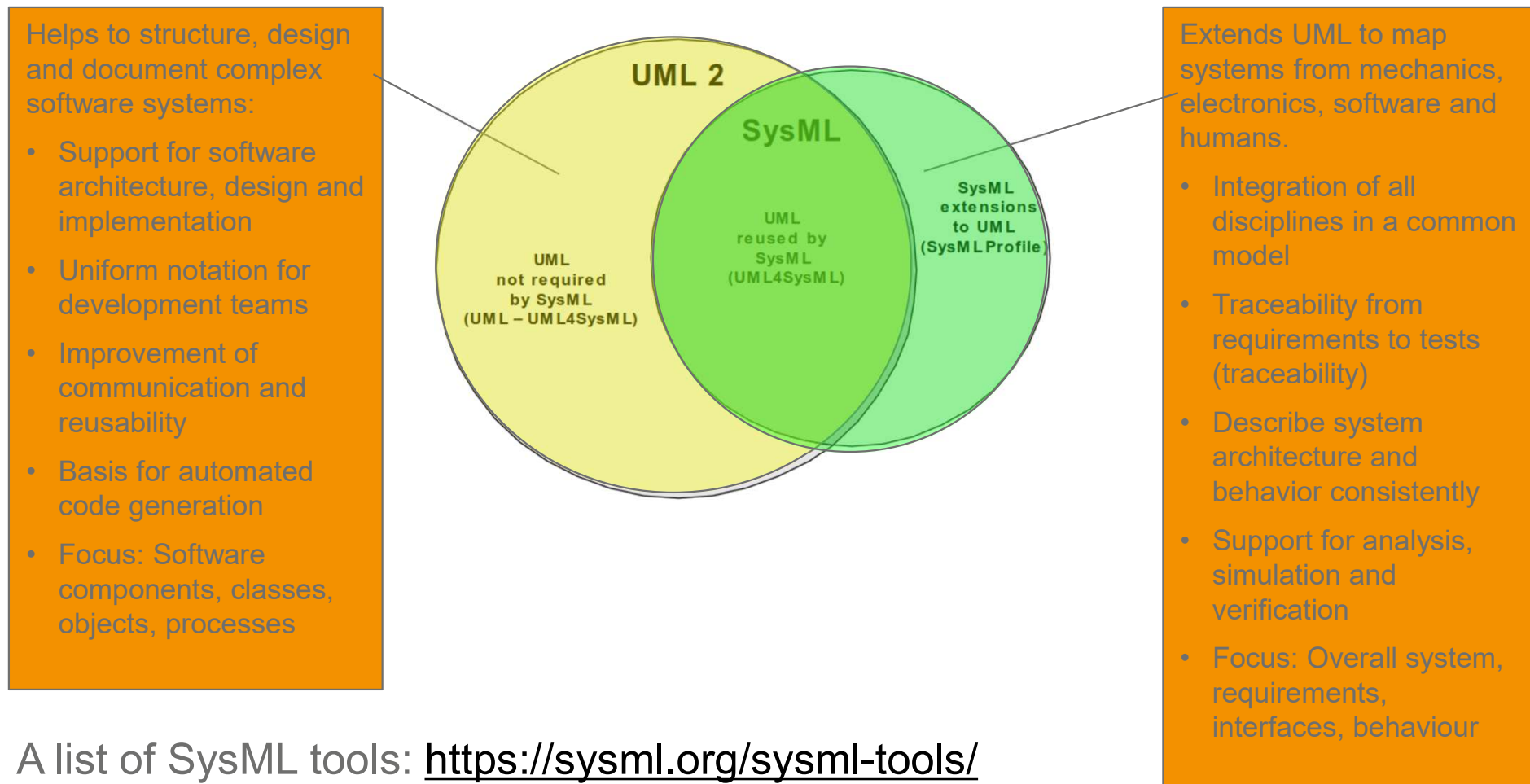
- Focus on OOP: Object-Oriented Systems Engineering Methods (**OOSEM**) (uses SysML to support the analysis, specification, design, and verification of systems)

<http://www.omgwiki.org/MBSE/doku.php?id=mbse:incoseoosem>

# Outlook - SysML



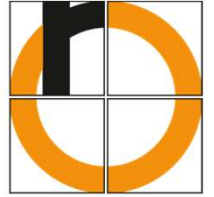
- SysML: System Modeling Language (<https://sysml.org/>)





**Simulation**





## Simulation purpose



Understand and reproduce  
a known phenomena.

Railway accident, Bad Aibling, 2016  
Tohoku Earthquake 2011

...



Optimize a known scenario

Transportation systems  
Flight plans  
Traffic light circuits

...

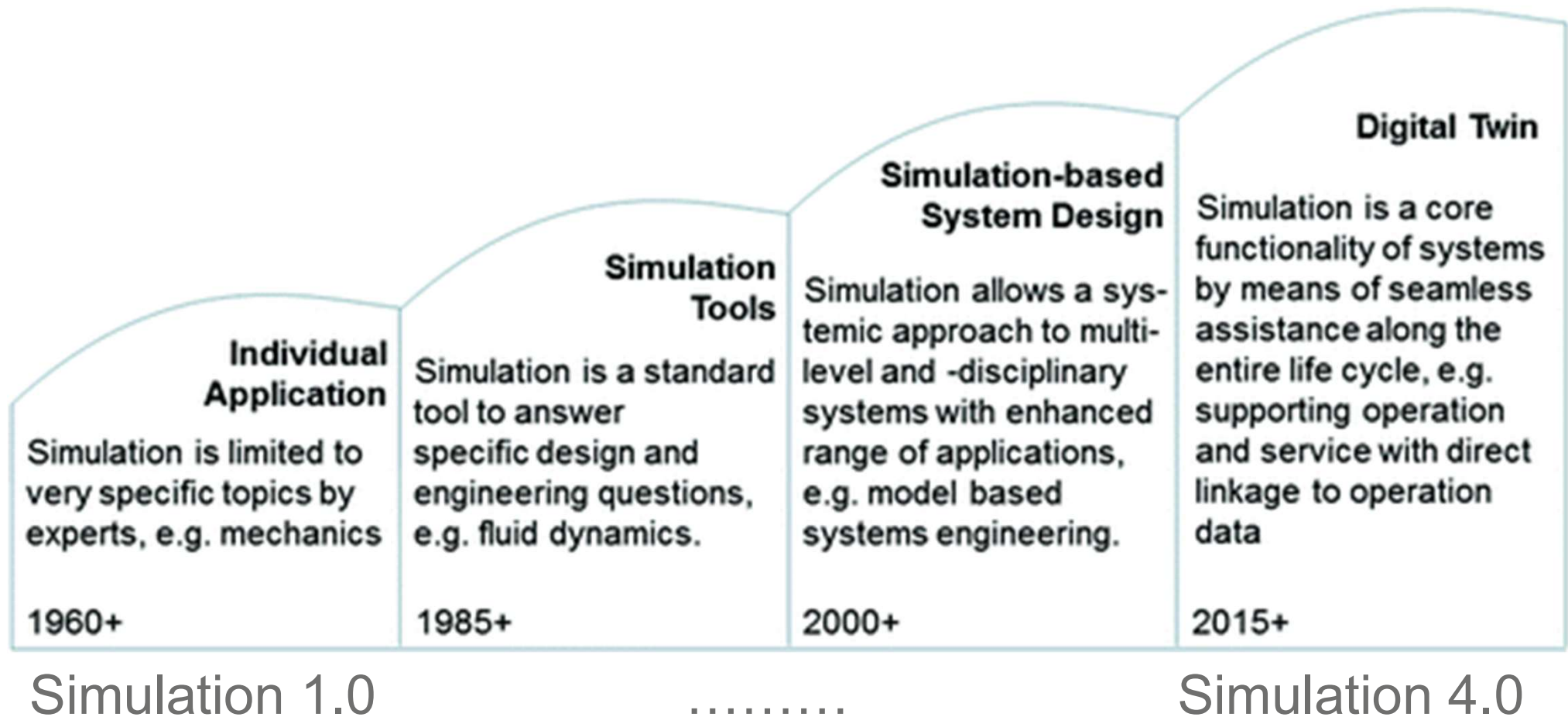
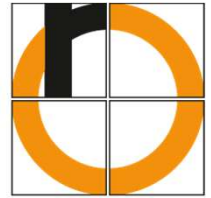


Predict a scenario

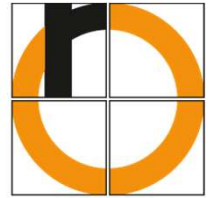
Weather  
Climate

...

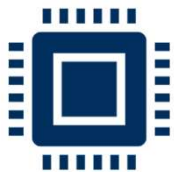
# Simulation evolution



Graph according to: Sobrino et. al., Digital Twin design at the material flow level; 2022 J. Phys.: Conf. Ser. 2212 012012; doi:10.1088/1742-6596/2212/1/012012



# Simulation types



## Physical simulation

A dedicated set of system attributes reproduced by physical models to achieve predictions with high accuracy

Can be costly → mockups



## Computer based simulation

Models of computation (MoC) of a discrete event or finite element → math. model as basis

Stochastic simulation – include i.e., Monte Carlo Simulation for statistical analysis



## Hardware / human in the loop simulations

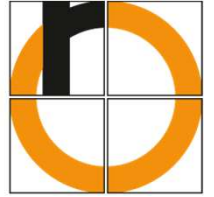
To test subparts of the system by applying i.e., a hardware to a simulating environment.

The simulating environment represents the real system providing simulated input to the hardware under test.

The hardware under test provides output that is fed back into the simulation environment.

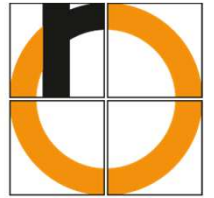
Mostly, this is a real-time simulation.

# Recap - System properties

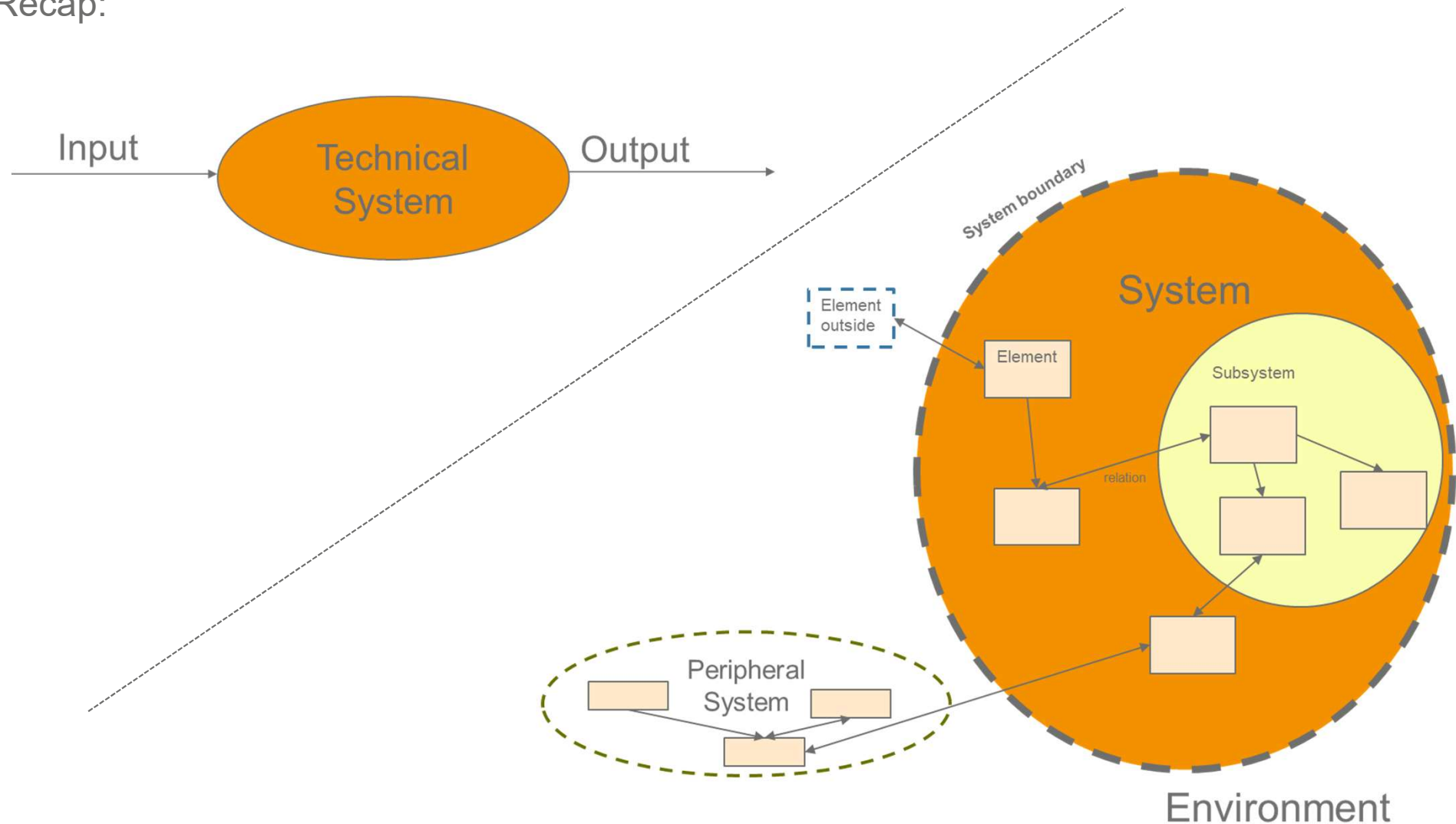


- A system exists in a broader context: it is **embedded in an environment** such as an operational environment.
- A system is composed of **several interacting components**. These include, for example, HW, SW, processes, system services, people, other organizations, and so on.
- A system has **emergent properties**: Properties that only arise from the system's interconnection.
- A system **goes through a life cycle** and fulfils certain functionality with certain behavior and performance.
- A system adapts to its environment; it must therefore be **able to change**.

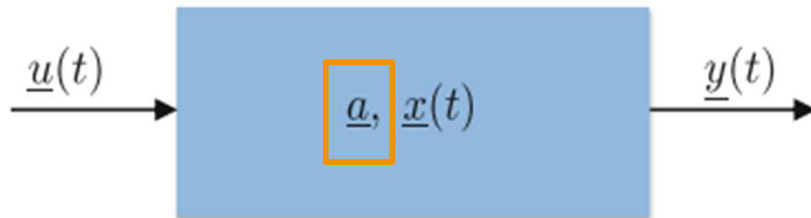
# Technical System / system within environment



- Recap:



- “ ... A technical system is characterized by the function of converting, transporting and/or storing substance (material), energy and/or information. ... “ (1)



Block representation of a system with inputs, outputs, state variables and parameters

$$\underline{a} = \begin{array}{ccc} \text{eingabe} & \text{datentyp} & \text{einheit (falls vorhanden)} \\ \left[ \begin{array}{l} ( 77.0 \\ ( \text{true} \\ ( \text{"Fälle Baum 27"} \end{array} \right. & \begin{array}{l} , \text{real} \\ , \text{boolean} \\ , \text{string} \end{array} & \begin{array}{l} , \text{m/s} \\ , - \\ , - \end{array} \end{array} \Bigg]$$

Example

- $\underline{a} \in V_a^{n_a}$  Vektor  $\underline{a}$  – **attributes / parameter of a system** with dimension  $n_a$ 
  - Examples: controller constants, duration of individual process steps, intensity values, etc.

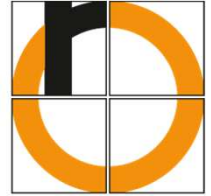
- i-th system parameter**  $\underline{a}_i = (\omega_{a,i}, d_{a,i}, l_{a,i}) \in V_a = (W_a \times D_a \times L_A)$

- $\omega_{a,i}$  : arbitrary values from the assigned range of values  $W_a$  (“real values”)
- $d_{a,i}$  : data type out of  $D_a$  (integer, float, boolean, ...)
- $l_{a,i}$  : given unit out of  $L_A$  (m/s, mm, ...)

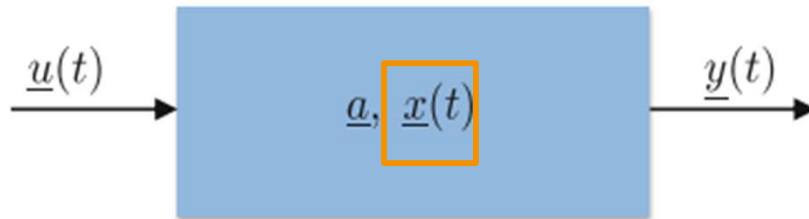
“Größen” (physical, technical, scalar)

They are related to an object of the system.

# Technical System



verschiedene zeitschritte  
a nimmt dadurch verschiedene werte an



Block representation of a system with inputs,  
outputs, state variables and parameters

Inputs and outputs of a system represented  
by  $\underline{u}(t)$  and  $\underline{y}(t)$

Behavior of the system: Progression of state variables over time

$$\underline{x}(t) \in V_a^{n_x} \text{ **State Vector** } \underline{x} \text{ of dimension } n_x$$

Current state means state at time  $t$

$$x_i = (\omega_{a,i}, d_{a,i}, l_{a,i}) \in V_a = (W_a \times D_a \times L_A) \text{ as **System state**}$$

- Examples: Location of body, joint angle, pressure, opening conditions of valves, etc.
- It takes on different values  $\rightarrow$  "**Größen**" (from previous slide)

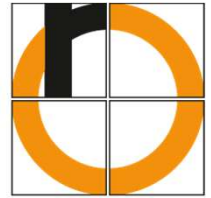


# Simulation model

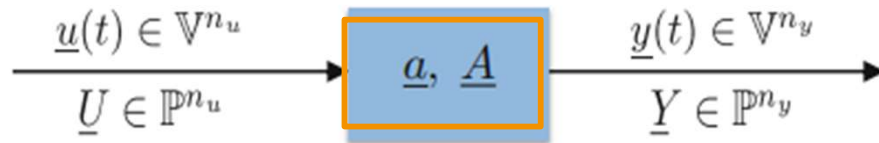
- **Modeling**: process of describing the structure, behaviour, properties of a system;
  - Product: Model
- „ ... **A simulation model is a formal, simulation-specific model of a planned or existing system. ...**“
- The **modeler creates** several simulation models using different simulation methods.
- The **correctness** of the simulation models is checked by verification and validation.
- It is **evaluated** whether the models depict the structure and behaviour of the real system accurately enough.
- Through **model integration**, application scenarios are realized by combining individual simulation models.
- How is a simulation model defined?

Source: Schluse et al., Experimentierbare Digitale Zwillinge, Springer Vieweg, 2024

# Simulation model



zusammenspiel mehrerer systeme



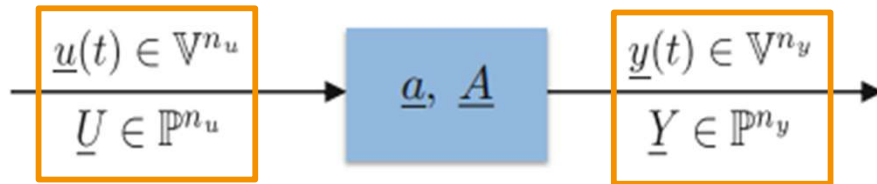
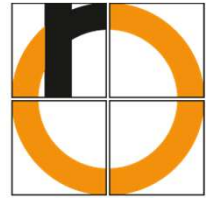
Block representation of a system with inputs, outputs, state variables and parameters

$$\underline{A} = \begin{bmatrix} ( \text{"while (t <= 5.0)"}, \text{MATLAB} ) \\ ( x_1(t) = v \cdot t + x_1(0), \text{Math} ) \\ ( \bigcirc \rightarrow \bigcirc, \text{Statechart} ) \end{bmatrix}$$

Beispiel Algorithmus-Vektor

- $\mathbf{M_S}$  is an element of the set of all **possible simulation models**
- Mathematical: ordered tuple  $M_S = (\underline{a}, \underline{A}, \underline{U}, \underline{Y}), \quad M_S \in \mathbb{M}_S := \mathbb{V}_a^{n_a} \times \mathbb{V}_A^{n_A} \times \mathbb{P}^{n_u} \times \mathbb{P}^{n_y}$ 
  - Vector of Model Parameters  $\underline{a} \in \mathbb{V}_a^{n_a}$  of dimension  $n_a$
  - Vector of algorithms  $\underline{A} \in \mathbb{V}_A^{n_A}$  of dimension  $n_A$  with indication of the selected simulator
  - $\underline{u}(t)$  und  $\underline{y}(t)$  --- Input and output (next slide)
- Algorithms model the behavior of components/systems.
  - Algorithms in the field of information processing systems
  - Algorithms for the explicit description of behavior of physical systems
  - Examples: State diagrams, Differential equations or Block diagrams

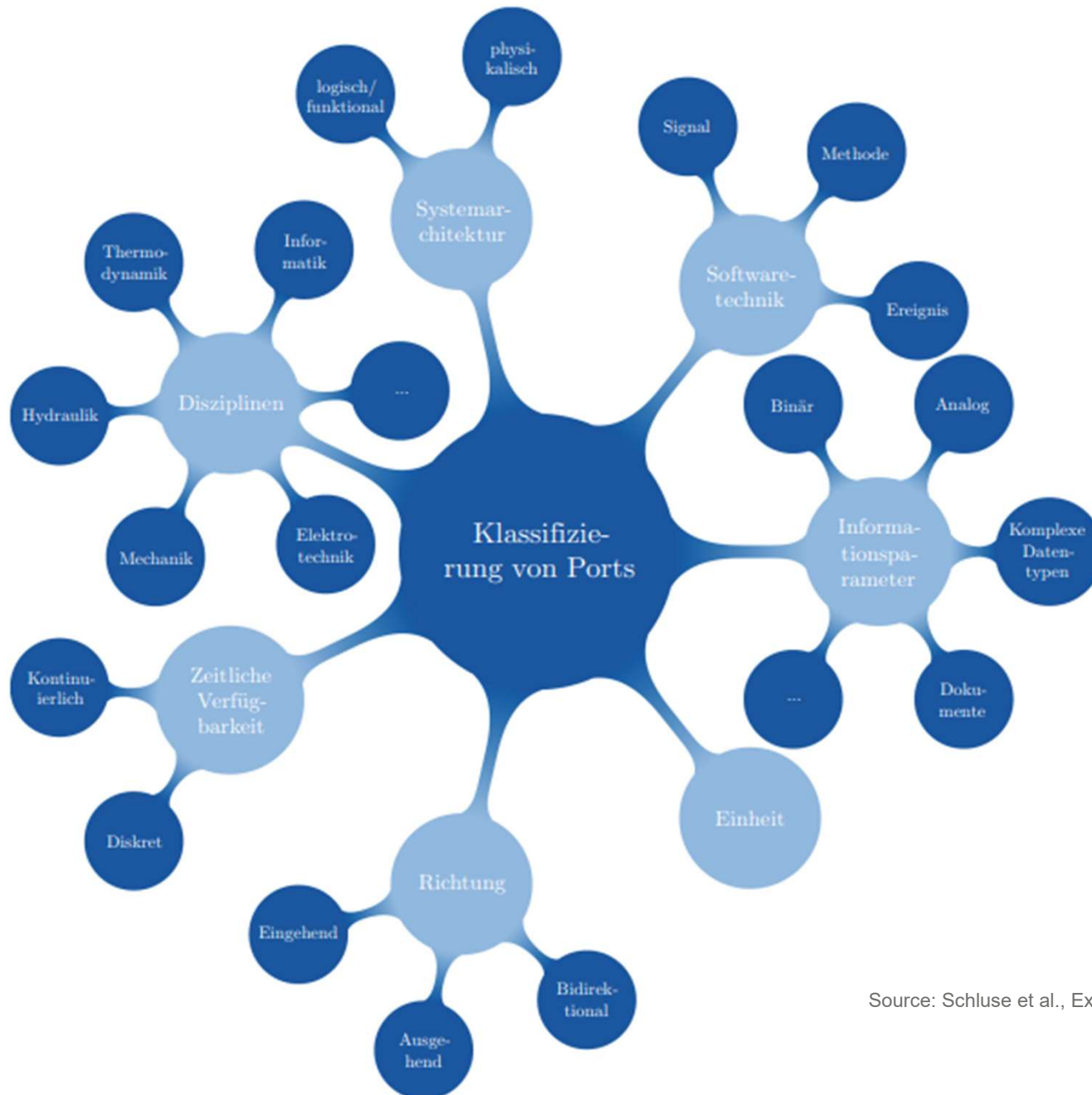
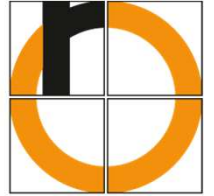
# System inputs and outputs



Block representation of a system with inputs, outputs, state variables and parameters

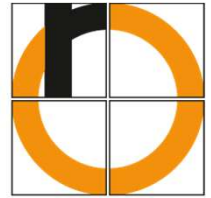
- $\underline{U} \in P^{n_u}$  and  $\underline{Y} \in P^{n_y}$  describe the **system inputs and outputs**.
- Any of these vectors can also be empty.
- They are also called "**ports**".
- $\underline{U}$  and  $\underline{Y}$  serve as a description; the concrete values are given by  $u(t)$  and  $y(t)$ .
- Port definition:
$$\underline{U}_i, \underline{Y}_i = (b_i, d_i, l_i) \in \mathbb{P} := \mathbb{B} \times \mathbb{D} \times \mathbb{L}_a$$
  - $b_i$  = unique identifier
  - $d_i$  = Data type
  - $l_i$  = Unit
- Ports can transmit specific values as well as absorb or forward algorithms.
- Directed arrows: describes causality (cause = input, effect = output)

# Classification of ports

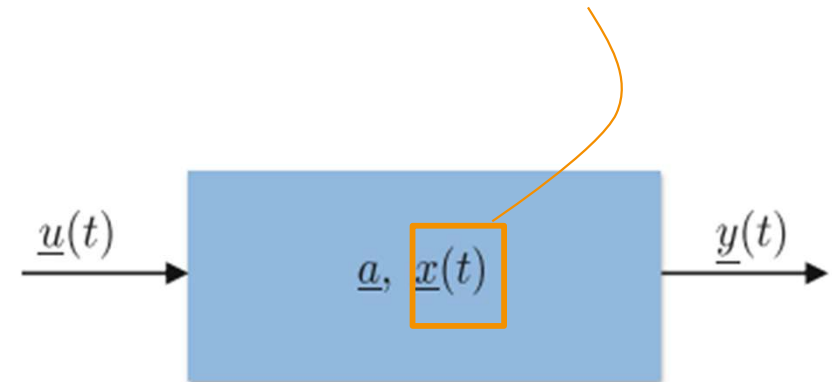


Source: Schluse et al., Experimentierbare Digitale Zwillinge, Springer Vieweg, 2024

# Experimentable model



- Mathematical: **Simulation calculates the temporal course of the state vector  $\underline{x}(t)$  with parameters  $\underline{a}$  and algorithms  $\underline{A}$ .**



- The **simulation model** and the parameters  $\underline{a}$  and algorithms  $\underline{A}$  used there can be time-dependent (e.g. change in its composition)
- What is a parameter for one algorithm (or simulation method) is system state for the next algorithm (or simulation method).

Source: Schluse et al., Experimentierbare Digitale Zwillinge, Springer Vieweg, 2024

# Simulation state

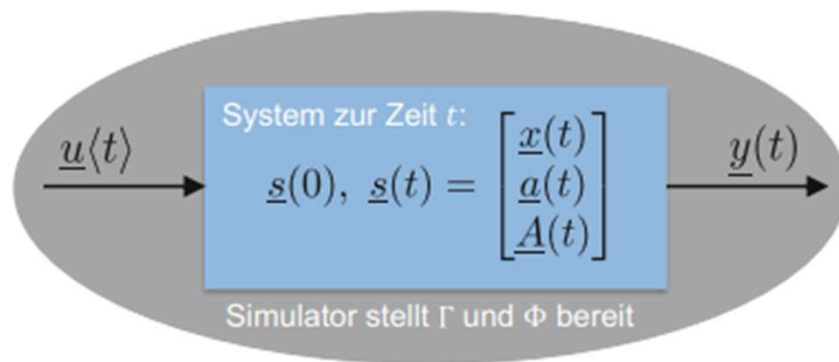
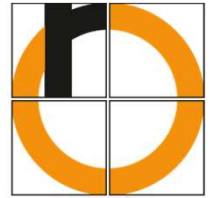
- „ ... In addition to the **system state  $x$  at time  $t$** , the **simulation state also includes the parameters  $a$  and algorithms  $A$**  of a simulation model for this point in time....“

$$\underline{s}(t) = \begin{bmatrix} \underline{x}(t) \\ \underline{a}(t) \\ \underline{A}(t) \end{bmatrix}, \quad s_l = (w_l, d_l, l_l) \in \mathbb{V} := \mathbb{W} \times \mathbb{D} \times \mathbb{L}_a, \quad \underline{s} \in \mathbb{V}^{n_s}$$

- **$s(t)$  - Simulation state vector**
  - Corresponds to the concatenation of state vector, parameter vector and algorithm vector

Source: Schluse et al., Experimentierbare Digitale Zwillinge, Springer Vieweg, 2024

# Simulation state



Block representation of a simulated system with inputs, outputs and internal state definitions

$\Gamma$  and  $\theta$  as simulation functions: mathematical representation of simulation model

$\Gamma$  : Simulation Transition Function

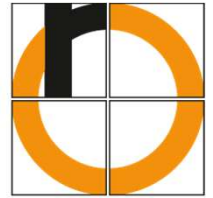
$\theta$ : Simulation output function

} are implemented in a simulator.

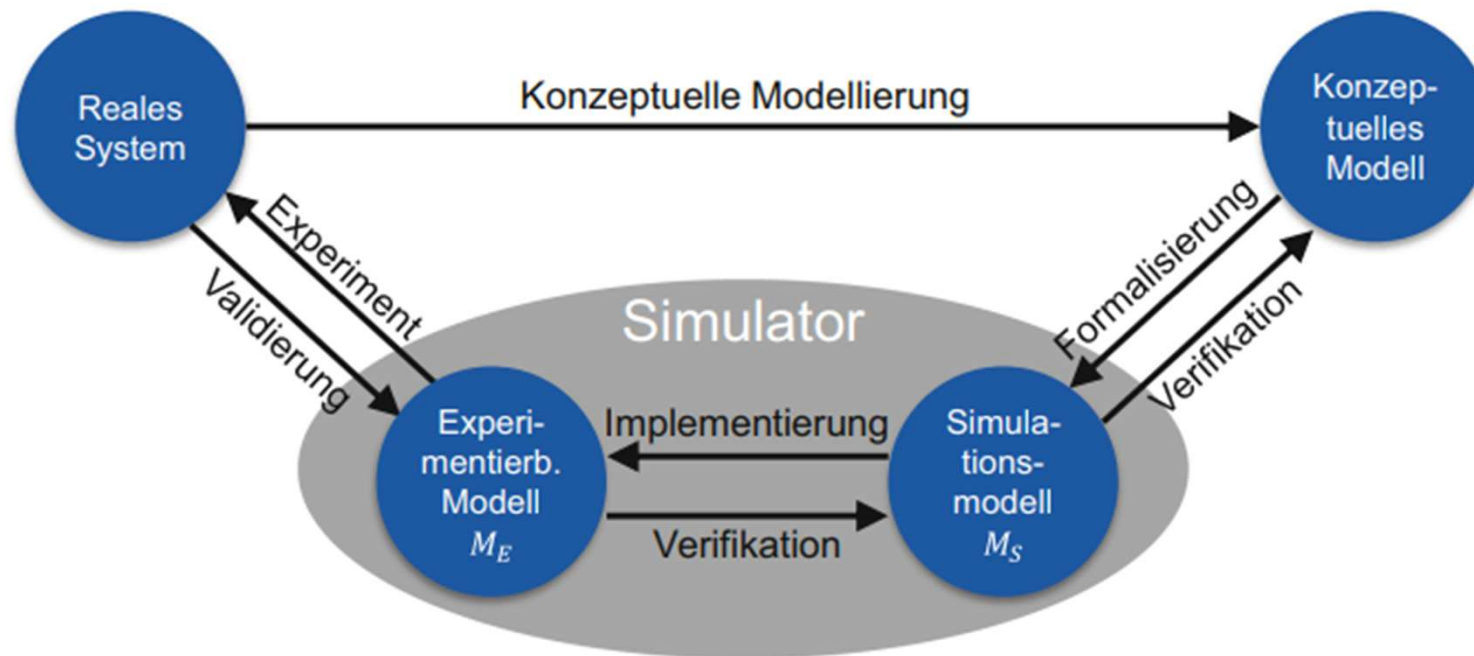
$\Gamma$  and  $\theta$  calculate based on the initial simulation state  $s(0)$  and known progression of the input variables  $u(t)$  the temporal course of the simulation state  $s(t)$  and the initial variables  $y(t)$



# Summary



- „ ... Ein konzeptuelles Modell ist eine software-unabhängige Beschreibung eines (bereits entwickelten, gerade entwickelten oder noch zu entwickelnden) Simulationsmodells und beschreibt dessen Ziele, Ein- und Ausgänge, Inhalte, Annahmen und Vereinfachungen. ...“

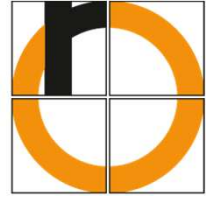


Source: Schluse et al.,  
Experimentierbare  
Digitale Zwillinge,  
Springer Vieweg, 2024

Illustration of the chosen relationships between the terms  
conceptual model, simulation model, experimentable model,  
model verification and model validation

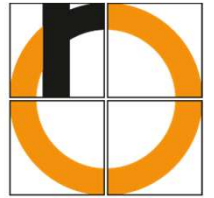


# Simulation Data Management



- Digital representation of the real system is model-based, supplemented by simulation.
- The real system and its digital twin is measured on quality, cost, and development time.
- Simulations help to support those factors:
  - Understand, optimize, and validate the product
  - Explore various solution spaces
- BUT:
  - Huge datasets
  - Long simulation times in complex projects
  - Simulations must be repeated – repeated data is produced
  - Data representation varies from tool to tool
  - Simulations aren't connected (“island-setup”)

# Why Simulation Data Management?



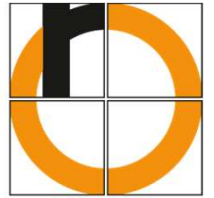
- Data should be exchangeable within various model/simulation approaches
- Data should be accessible through various steps of the modeling and simulation chain.
- Data should be managed.
- Data should not be repeatedly produced – provide a mechanism to avoid duplicate data.



# Key terms in this context

- **Product Lifecycle Management (PLM)** = efficient management of data within whole product life cycle.
  - control product along life cycle
  - improve processes
- **Product Data Management (PDM)** = techniques for gathering, classifying and storing data.
- **Simulation Data Management (SDM)** = manage **data** resulting from simulations; synonym: engineering data management
- **Simulation lifecycle management (SLM)** = manage **processes** within simulations
  - Coherent knowledge from process to product data
  - Organize, structure, store and track produced information

# Simulation data categories

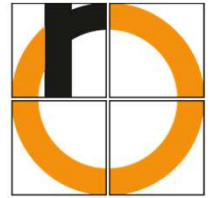


- **Short term** – data produced by solvers from which “relevant” information is extracted → typically deleted after some time.
- **Long term** – data produced by analysis and interpretation to defend the performance of the underlying system and to communicate to the stakeholder. Also contains the simulation specification, such as underlying method, parameters, etc.
- SLM key objectives:
  - Collaborative product development
  - Data traceability
  - Decision support
  - Simulation system integration
  - Process automation



# Simulation Process

Time consuming,  
problem space  
analysis is cost-  
intensive!



## Modeling

- Design system of interest – system components
- Define problem space – phenomena
- Consider Boundary conditions
- Pre-Processing = discretization (space,time,..)

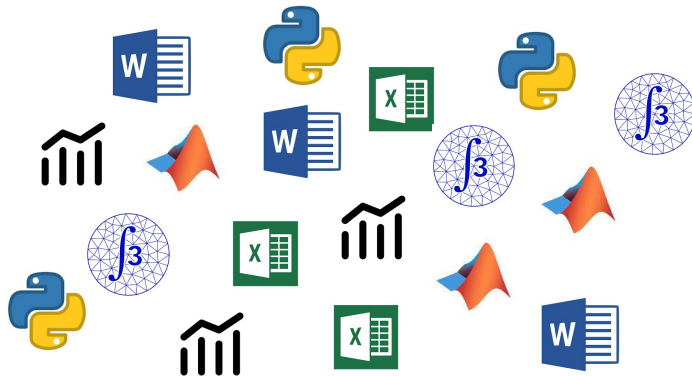
## Solving/Simulation

- Based on algorithms
- Numerically, analytically
- Vary in time

## Analyzing

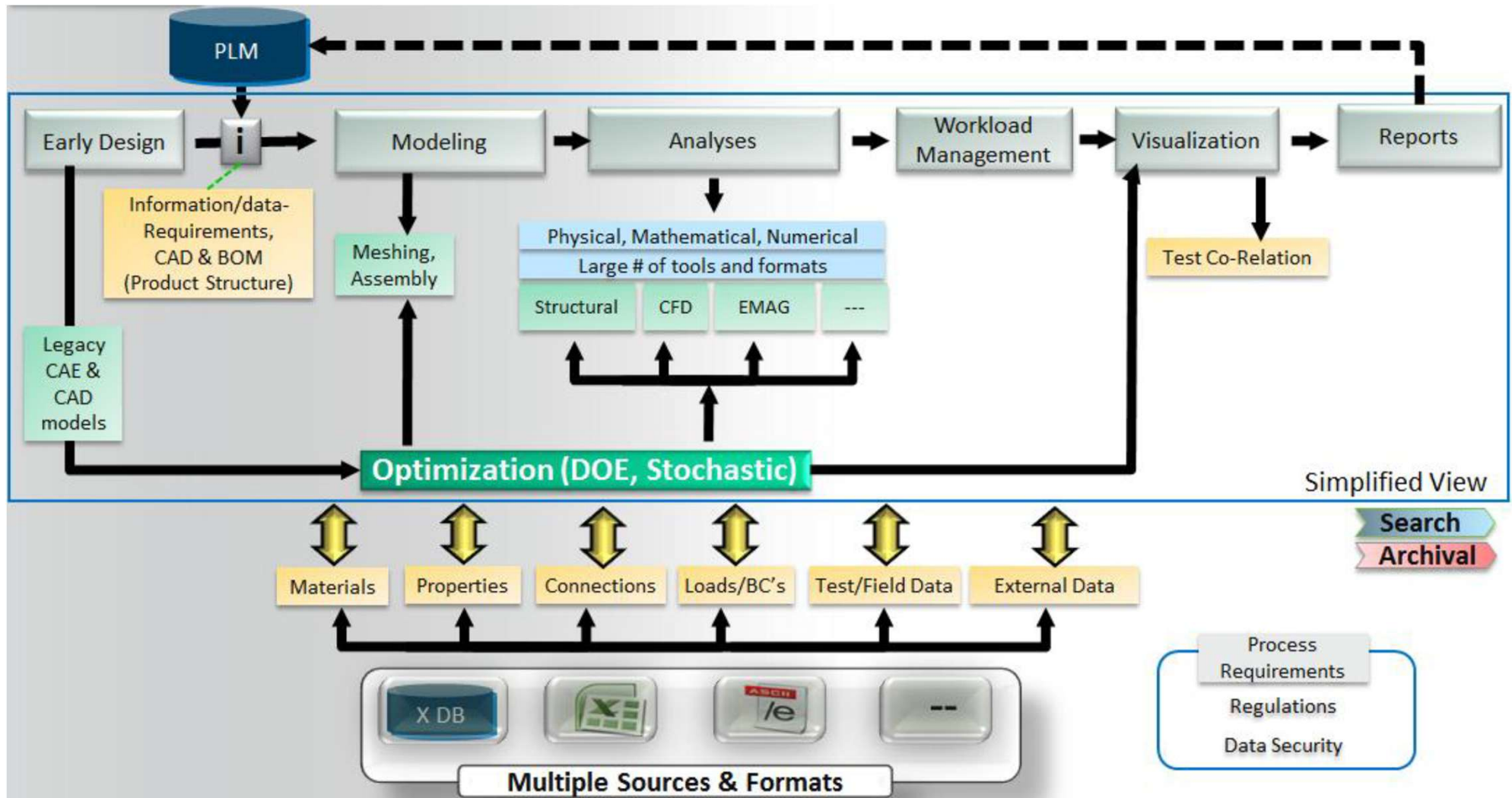
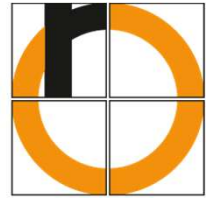
- Validity of results
- Statistics
- Trade space analysis
- Intuitive representation
- Comparison against stakeholder needs

Resulting in data, data, data (of any kind)



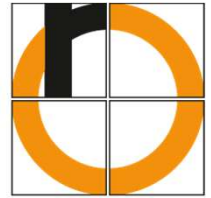


# Sources of Engineering Simulation Data

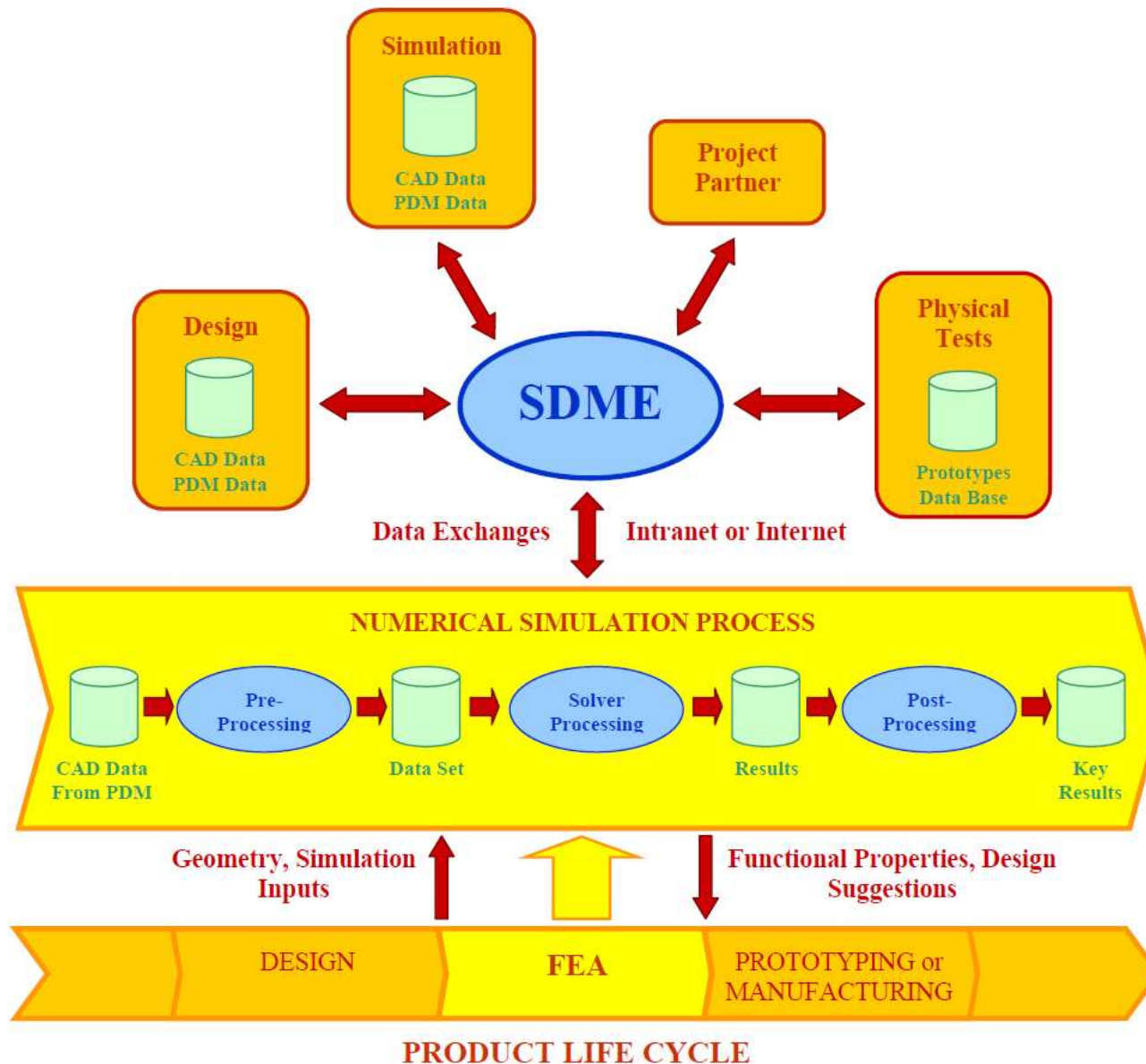


S. Chari, Addressing Engineering Simulation Data Management (SDM) Challenges, White Paper, Cabot Partners, 2013

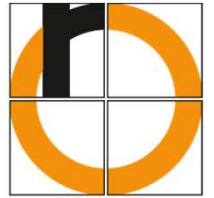
# Position in Product Life Cycle



Source: Charles et al.;  
Specification of simulation  
data management  
environment integrated with  
PDM. International  
Conference on Engineering  
Design ICED 05 Melbourne,  
2005



# Simulation Data Management Challenges (according to [2] )



- Growing volume, velocity and variety of simulation data
- Maintaining data local to HPC resources
- Managing isolated distributed data
- Ensuring security and integrity
- Improving collaboration and communication
- Enhancing knowledge capture
- Integration with other systems
- User-centric data management

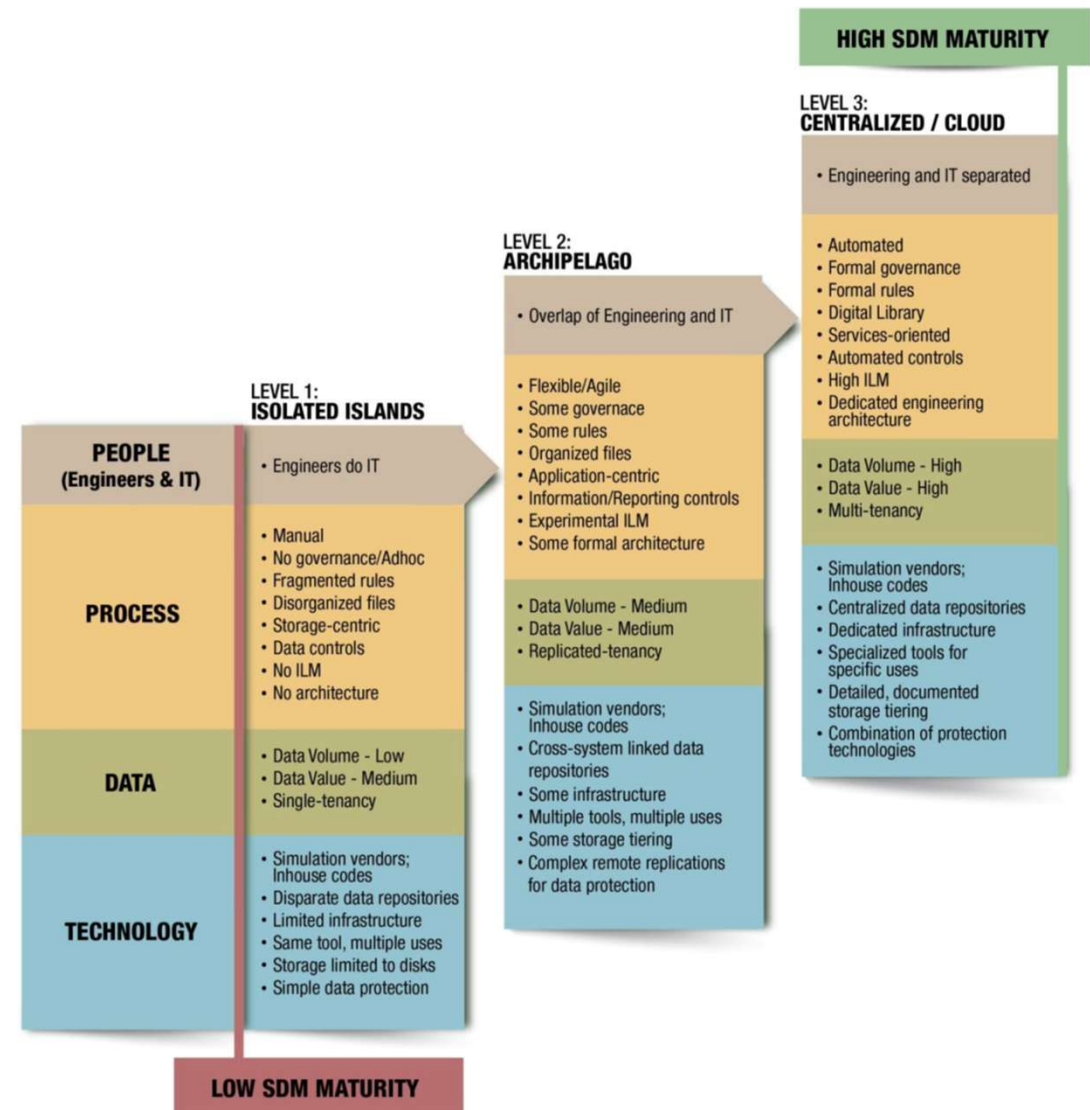
Source: S. Chari, Addressing Engineering Simulation Data Management (SDM) Challenges, White Paper, Cabot Partners, 2013

# SDM Maturity Levels in companies (according to [1] and [2])



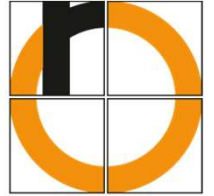
## SDM Maturity Levels

- Key goal: efficient data structuration
- Potentials:
  - Fast data access from anywhere
  - Access for each stakeholder
  - Enhanced decision support
  - Data re-using
  - Automated reporting
  - Improved productivity
  - Centralized and secured database
- Challenges:
  - Collect store and retrieve growing volume, velocity and variety of simulation data (big data)
  - Manage network
  - Improve collaboration and communication
  - Ensure data security, integrity
  - Data management system integration



- (1) Blondet et al.; Simulation data management for adaptive design of experiments: A literature review; Mechanics & Industry 16, 611 (2015)
- (2) S. Chari, Addressing Engineering Simulation Data Management (SDM) Challenges, White Paper, Cabot Partners, 2013

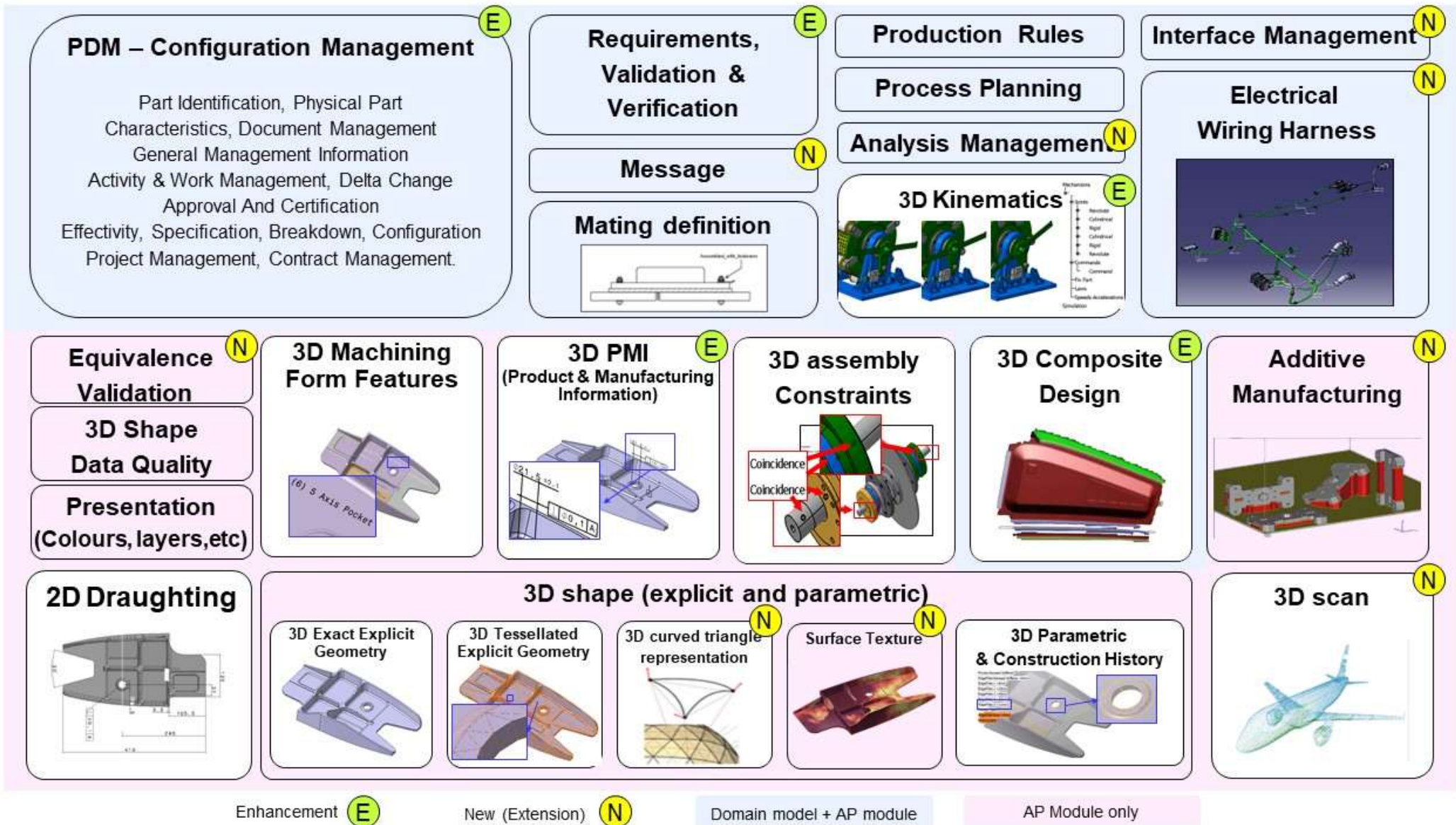
# A Common Data Model - Standardization



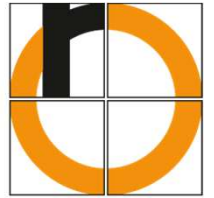
- Goal: Enhance data sharing and Integrate multiple data sources
- → Provide a “neutral format”
- STEP: “Standard for Exchange of Product data-model Family”
  - **AP209:** “Multidisciplinary Analysis and Design”, (<http://www.ap209.org/>), 2001
    - Data model to share CAE data during product life cycle
    - Neutral file format to enable interoperability between different systems
    - Version control
    - Supports simulations
    - But: lack of integrated processes
  - **AP242:** “Managed Model Based 3D Engineering”, (<http://www.ap242.org/>), 2014
    - Data model to manage multi-disciplinary design processes
    - Ensure linkage between models and processes
    - Integration of simulation data management systems



# ... Standard für die modell-basierte Produktentwicklung ... STEP AP 242

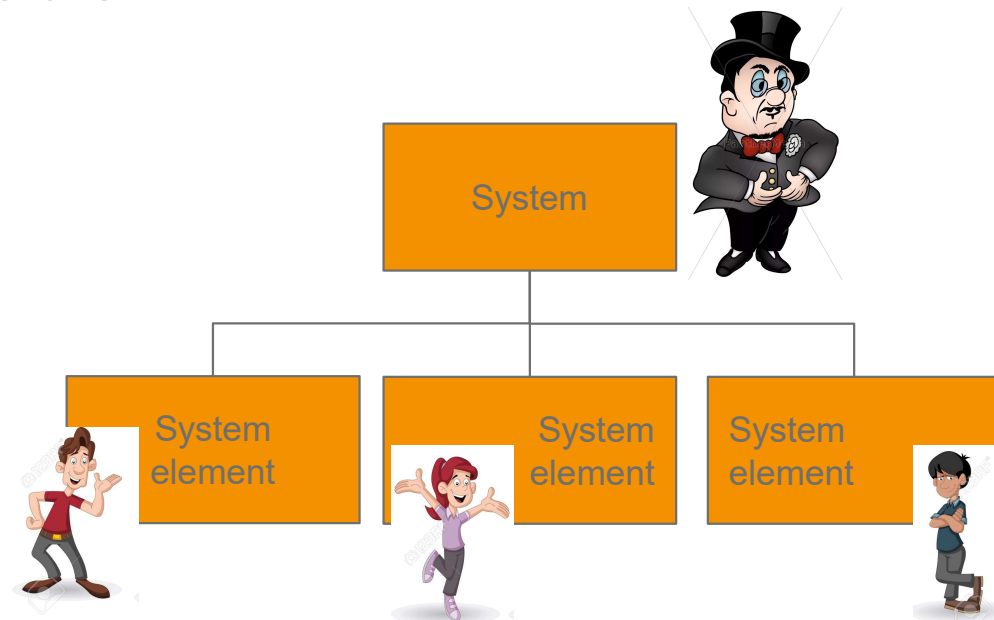


# Simulation Model Management?

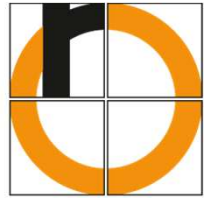


- Who is the responsible person within a model centric environment?
- Who takes ownership?
- Who is acting as moderator and coordinates changes?
- Who is paying for all this and who is taking care that there is a budget available?

- Idea:

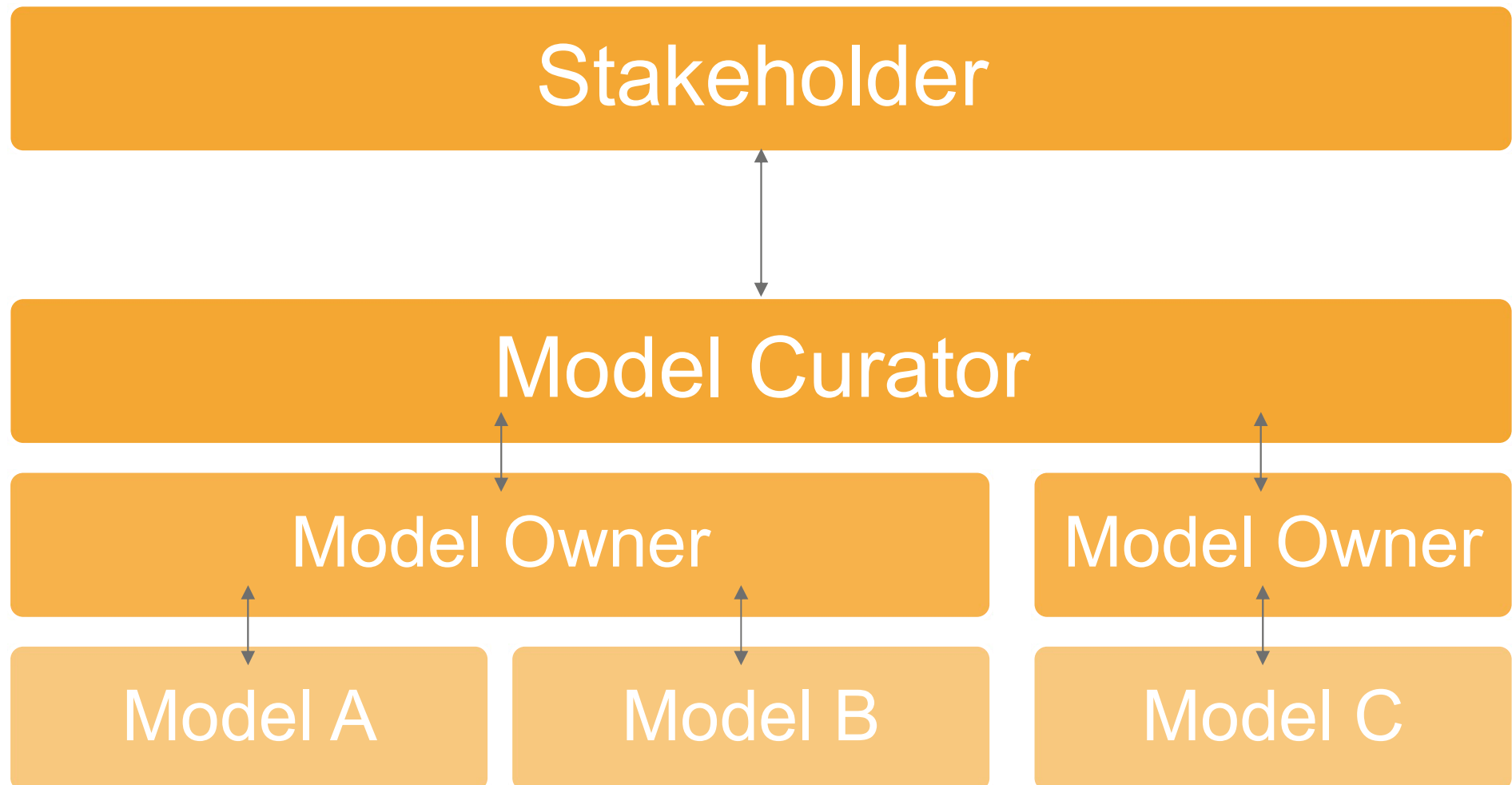
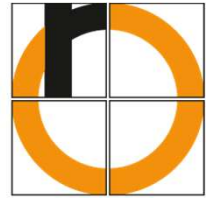






- Introduction of a Model Management Plan (MMP):
  - Initial verification and validation of the models
  - Governing the inputs to the models
  - Communicating model results
  - Model configuration management
  - Defining the model owner
  - Funding the model
  - Process for model changes over the lifecycle
  - Model end of life and renewal planning

Fisher, Friedenthal, Sampson, VanZandt, Palmer, Nolan, Loeffler,  
Bajaj, Hovey, Hart; Model Lifecycle Management for MBSE:  
<http://www.omgwiki.org/>



Reymondet et al; "Considerations for Model Curation in Model-Centric Systems Engineering," 10th Annual IEEE Systems Conference, Orlando, FL, April 2016.

# Time for questions ...

